

A Literary Essay

The Inner Horizon

Chamber Edition

Íñigo Barrera Barceló

*To Montse and Gerard,
for putting up with me every day with a smile.*

Contents

OVERTURE — The Habit of Water	4
I — The Shape of the Interior	10
II — The Time of the Bond	20
III — The Workshop of the Bond	26
INTERLUDE — The Edge We Cross Every Night	39
IV — The Wounded Horizon	44
V — The Geography of What's Missing	52
VI — The Limit	56
Author's Note	60
Intimate Glossary	62
Notes and Sources, Chapter by Chapter	64
CODA — Txiki	68
The Mirror Without Depth	69
The Invisible Tuning Fork	91
The Eye of a Single Color	106
The Fractal Reality	118



OVERTURE — The Habit of Water

The water began to withdraw without ceremony. There was no warning, no bell. In Tarel, speech always came late: things happened first, then stayed, and in time became habit.

The city floated on a brackish lagoon so old that no one remembered a beginning. Wooden houses, worn ropes, joints made with patience. Children learned the sway before they learned the names of the streets; they knew how to come home before they knew how to leave. Coming home was a verb: a repeated action, a small habit of the body.

The water was not a backdrop or a landscape. It was the habit itself.

That is why the change did not arrive as absence, but as strangeness. The oar touched bottom where before there had been depth. The boats woke up crooked, as if they had spent the whole night doubting. The nets came back light; the faces, heavy. The silence widened, as if something knew the moment had come to let go.

I was the archivist. Or so my appointment said. In truth I watched, slowly, whatever did not yet ask for a name: minor tides, births without ritual, deaths without announcement, stories the old people told to the wall when the world pretended not to listen.

I noted the first strip of mud without urgency, the way one corrects an old map without yet knowing what city will remain on the far side of the ink. I got the first figure wrong: I wrote that the water had dropped a handspan, and it was two. It took me a week to notice, and by then the

wrong dates had already been copied onto three different pages. I did not correct them all. Some stayed wrong, and remain so.

"It wasn't there before," said Kema, pointing to the dark edge beneath her house. "The water covered it. The water knows."

"Knows what?" I asked.

"When to let go... and when not to."

I wrote it down, though I did not know under what heading to file it.

The water withdrew without hurry. It did not tear away: it took its leave. It bared rotten piers, mooring rings crusted in mud, and marks of past levels that no one had wanted to read as a warning.

It was a guest who packs while you sleep and leaves the room ready for someone who will not come.

Two stories lived in the archive. One said that Tarel had always floated. The other, that we had fled to the water after a war, or after a thirst so old it no longer hurt. In both, the same word appeared, written in different hands: home.

Months later the delegation from the Dry Continent arrived. Their boats were rigid. Their clothes did not learn to get wet. They walked slowly, as if the ground might answer them. They asked about everything. They listened less than they spoke. They measured the lagoon in silence and spread out papers the wind tried to fold.

They gave figures without raising their eyes. They said the level would fall further. They did not say for how long, nor at what cost. Someone asked if the water would return. They said they did not know, and said it in the same voice one uses to say it will not rain.

"You will have to choose," they said. "Stay or leave."

No one answered right away. I noted the phrase the way one notes the weather: too general to argue with, too precise to ignore.

Tarel split without breaking. Some families dismantled their houses with an efficiency that was frightening. They burned dry algae. They promised to return. They promised to write. Not everyone looked back as they left.

Others stayed and learned the speech of the mud: where to step without sinking, when to wait, which sound announced a crack and which only weariness. Some stayed and adjusted their footing; others kept treading the same print.

The children walked. Some mourned the lost floating. Others showed off new blisters like medals no one had given them.

The archive changed. It no longer held only stories of the water, but of what remained when the water left. Seeds appeared beneath the silt, rusted tools, marks we did not know whether to read as past or as future.

I found an unsigned text, nearly erased by salt: "The water sustains us as long as we remember that we sustain it too."

I dreamed of the drained lagoon only once. It was not dead. It slept beneath layers of earth. It breathed slowly. When I woke my hands were closed, as if they still held something.

The last message from the Council arrived. They recommended evacuation. They used a word that sounded exact and did not fit the language of Tarel. We decided not to repeat it. Not because it was false, but because it did not know where to live.

There was no rebellion, no ceremony. There were small gestures: new ropes strung between platforms and solid ground, houses that leaned

before accepting the weight, old songs sung more slowly to accompany footsteps where before there had been oars.

I left gaps in the archive. There are changes that admit no full record.

On the last day I put my hands into the lagoon that remained. It was warm, murky, salty. I let it run between my fingers. I did not say goodbye.

The water of Tarel returned at night. Without announcement. Without signs.

In the morning the people found the shore where it had always been. The same mud. The same stones. The same line of salt on the low walls.

The papers of the Dry Continent still sat in the archive, their exact figures and their ink already pale. They measured a lagoon that had stopped needing to be measured. No one consulted them again. The water does not read maps.

No one could say whether the water had brought something with it. No one could say whether it had left something behind.

Poem: The Archivist

*What leaves
leaves the door ready
for another name.
I note the gap,
not what was there.
The letter does not know
it is signing a farewell.
The children trade
their blisters for coins
of a country not yet minted.
I keep the blanks in the notebook.
Someone will read them
the way one reads a silence
well written.*

Poem: The Oar

*The oar touches bottom
where once there was depth.
The water fell less
than the blow rose
through my arm:
a warning that emptiness weighs.
I have seen something surface from my own reservoir.
I have seen it return.
I repeat the gesture
to make sure,
each time,
that there is still a bottom.*

Poem: The Shore

*And the water came back to the shore
without warning.
I could not tell — not as archivist of my own cracks,
nor as the child I was—
whether it was the same shore
or a new one
with the same name.
No one could tell what it brought,
nor what it carried off.
But at dawn
I looked at my open hands.
Perhaps
that was the answer.*



I — The Shape of the Interior

Essay: Three Images and a Boundary

When the water began to withdraw from Tarel, its people sensed that some elemental form of boundary was closing. I understood much later that this same intuition governs our own center. This book is, at bottom, the exploration of that hunch carried to its final consequence. To enter this territory, one does not need quantum physics; three images that our everyday experience already possesses will do.

The first is the soap bubble. As you blow, the soap stretches, curves, and closes upon itself: suddenly there is an inside and an outside that a moment before did not exist. What makes the bubble what it is is the membrane that encloses it, not the trapped air. The second is the ocean, a deep mass from which waves emerge to take on individual shape before being reabsorbed into that same surface. The third is the network, whose nodes only make sense through the threads that bind them: an integrated network is more than the sum of its loose nodes, and to that difference — how much the organization adds over and above the parts — the physics of information has given a name, Phi (Φ). The greater a system's Phi, the more it behaves as an inside and not as a mere collection of pieces.

The thesis reached me late, and it did not console me: consciousness is the interior of an encapsulated informational domain. There is a boundary separating two regimes of access, not two substances. On one side, everything that can be measured from outside: firing neurons,

neurotransmitters, words. On the other, what it feels like to process it from within: the red of red, the texture of an exact thought. Within this map, nothing crosses that boundary.

This is not an idle claim. One could know every last neural firing, every molecule, every connection in someone else's brain, and still not know what it is to see red from inside those eyes — just as one could memorize the entire physics of color without ever having seen red. That margin which no external description closes, however complete it may be, is the boundary. It is not that we have simply failed to cross it yet for lack of better instruments: crossing it does not belong to the class of things an increase in information can resolve. Data adds precision on one side of the boundary; it never carries anything across to the other.

Thought is not found under the microscope because that instrument is designed only to observe the public interface. The private state is, by construction, on the other side of the boundary. Asking an electrode to capture a subjective experience is like asking an observer outside a black hole to read a letter that has fallen inside it: the information is real, but the vantage point is geometrically wrong.

Encapsulation is the price a system pays for its own complexity — no biological accident at all. To process without dissolving into the noise of its surroundings, it needs to close itself behind a clean interface. An interior can only emerge, and sustain itself over time, if something separates it.

Poem: The Bubble

*The soap stretches between two hands
and suddenly there is an inside
that did not exist before.
I am the part of the orchestra
that fits inside no musician.
When it ends, I do not break.
I take my leave.
Something stays floating an instant,
ownerless,
before bursting in the still air.*

Fiction: The Thief of Interiors

They said of him that he could open anything.

It was simple fact, not another's boast. Doors sealed for generations, chests whose owners had died without opening them, boxes within boxes within boxes... everything yielded sooner or later to his hands, which knew how to listen for what a mechanism wants before the mechanism itself knew it was saying it. He did not steal for greed, but because he could not bear for something to be closed while he remained outside.

One day he was told of a box at the end of a house at the end of a city at the end of a road. It was said that inside lay a treasure no thief had ever managed to reach, not because the box was stronger than the others, but because it had seven locks, each of a different kind, and none opened with what had opened the one before.

He decided to go.

I. The First Lock

The first lock was of force.

It was the one everyone expected, the one any thief knows how to force: a mechanism of metal and resistance that gives way when more pressure is applied than it was built to bear. The thief studied it one night, set his lever at the exact angle, and the lock surrendered with a dry click, the sound of things that only know how to say no until someone insists enough.

He smiled. If they were all like this, it would be a short night. But, as he suspected, they were not all like this.

II. The Second Lock

The second lock was of patience.

There was nothing to force, quite the opposite: it was a mechanism that opened only if turned slowly, so slowly that any haste jammed it again from the start. The thief, used to quick skill, had to learn something else. He spent three days turning a wheel one degree an hour, undoing with a single tremor an afternoon's work and starting over, until on the third day his hands learned not to want to arrive. The lock opened when he stopped being in a hurry to open it.

It was different from the first, which had asked of him what he had in abundance. This one asked something he did not know he lacked.

III. The Third Lock

The third lock was of silence.

It jammed at noise. Any sound — the scrape of a tool, his own breath, his heartbeat if he listened to it too closely — closed it. The thief had to still himself as he had never needed to before. He learned to breathe without the breathing being heard, and to be present without making

the noise of presence. And in that silence, for the first time in years, he heard things his trade had always drowned out: that his hands trembled even when he himself did not, that an old weariness lay beneath his skill. The lock opened when there was, around his hands, no one left to make noise.

IV. The Fourth Lock

The fourth lock was of renunciation.

It held a trap, simple and cruel: it carried two bolts that had to be opened at once, and he had only two hands, so to reach the second he had to let go of the tool with which he had opened the three locks before it. His father's tool, the one that had made him a thief. He had to let it fall into the shaft that opened beneath the box, with no guarantee that anything lay on the far side of the gesture.

He held it a long while, and at last let it go. He heard it fall, and heard that it never struck bottom, and the lock opened with the hand now free.

V. The Fifth Lock

The fifth lock was of memory.

It asked for a name. Not just any name: the mechanism yielded only to the name of someone the thief had loved and had left behind for his trade. He had several to choose from, and chose the one that weighed on him most. He said it aloud in that empty house, and hearing himself say it was worse than letting go of the tool, because the tool simply fell, while the name, instead, came back.

The lock opened, but he stayed a while with the name in his mouth, which did not close as easily as the bolt.

VI. The Sixth Lock

The sixth lock had no keyhole.

The thief searched it over: no slot, no wheel, no bolt, nothing his hands knew how to question. He spent a whole day running his fingers over a smooth surface that offered no way in. And little by little he understood that this lock did not open from outside: that the only way to defeat it was to stop trying to open it, to accept that there was a part of that box not reached by force, and to pass through it without forcing, trusting. He drew back his hands and stood still before the smoothness.

Only one remained.

VII. The Seventh and Last Lock

The seventh lock opened with the bare hand.

It asked for no tool, no patience, none of what the others had demanded. It asked only that he lay his open palm upon it and wait. He set his bare palm on the wood — the same palm that had forced so many doors — and waited. He felt, beneath the skin, something beating as living things beat, and for an instant, before the end, he was certain that on the far side of that last panel there was no object: that whatever the house had kept behind seven locks was not a thing that could be taken out and carried away.

So he pressed his palm, and the lock gave way, and the box vanished. It did not open: it vanished. In the exact instant the last boundary surrendered, the whole box — the metal, the seven defeated locks, whatever had been inside — ceased to be, as if it never had been. No gap remained in the air, no mark on the table, nothing left to steal, because there was no longer an inside to take it from.

The thief stood with his palm open over the place where the box had been. He did not cry out, did not curse: he stood very still, just as he had

a few minutes before, facing the smoothness, and understood — without words, in the way one understands things that cannot be said — why it had vanished.

The treasure had not been inside the box. The treasure was the inside itself.

The thief who could open anything had, at last, opened everything, and in opening it completely, had made it disappear.

He lowered his hand and walked out of the house at the end of the city at the end of the road with empty hands, which in some cities is the only honest way to come back from certain places.

He had stolen nothing, but he had understood something, which is the opposite of stealing: you do not take away a thing that belonged to someone else, you come away with something you cannot show to anyone.

Fiction: The Boy Who Learned to Measure Time

I am eleven years old and this summer never ends.

It is not that the days are longer. My father says June has the longest days of the year, and then they shorten, and he says it every June as if it were news. It is something else. Inside each afternoon there is room for more afternoons than fit in one afternoon. I do not know how to explain it. I only know how to tell it.

I have a notebook under my mattress, because if my sister finds it she will laugh. Inside I write down times that are not clock-times: shadow-time, ant-time, waiting-for-something-to-happen-time. I do not mix them up. Today's ant-time was long, because the ant was carrying something white and took a long while deciding where to climb the curb.

Mr. Montserrat lives next door; he has an olive tree that covers the whole yard, and a wooden chair where he sits with a book he almost never reads. Today I asked him how long summer lasts.

"Ninety days," he said.

"No. How long it lasts."

He looked at me for a long while.

"For me, less than before."

It was not the answer I wanted. But something tells me it is the right one.

He taught me a game with no name: sit under the olive tree, close your eyes, wait for something to happen. Nothing in particular. I wanted a measurable answer — how long do we wait? — and he said "as long as it takes," which annoyed me a little, but I sat down anyway, on the ground, which today smells of dry earth.

A bird landed on the lowest branch. I heard it move, its small weight making the wood creak. I did not see it — my eyes were closed — but I heard it move for a long while, and then I heard it leave, and then there was silence, and the silence lasted too.

"How long did it last?" I asked him.

"How long did it seem to you?"

"A long time."

"Then it lasted a long time."

Again, not the answer I wanted. Again, the right one.

Today a storm arrived that was not supposed to: clouds all at once at five o'clock, when there were still three hours of daylight left. Mr.

Montserrat did not move from his chair. He watched the olive tree move its branches in a way I had never seen before, as if the tree too knew what was coming. We both got wet. It did not seem to bother him. It did not bother me either, by that point.

It lasted ten minutes. Then the sun came back as if nothing had happened, just as Mr. Montserrat seemed to have known all along. I did not write it in the notebook. There are things kept elsewhere.

Mr. Montserrat died this morning. My mother told me without warning, the two of us sitting in the kitchen. I think he already knew. Those who spend their afternoons watching the sky with that calm tend to know these things before anyone tells them.

The olive tree is still there. That is how it goes with olive trees.

This afternoon I went to sit beneath it, alone, eyes closed, waiting for something to happen. I did not know what. A long while passed with nothing happening, and then the wind moved a branch, and I thought that maybe that counted.

I did not write it down. But I kept it.

More than thirty years have passed. I asked my mother again, over an August lunch that ran long, the iced coffee sweating in the glass.

"How long did summer last when I was eleven?"

She sat thinking.

"Three months," she said. "Like every year."

"No. How long it lasted."

She looked at me the way one looks at a son who has been asking the same question for thirty years.

"Whatever you wanted," she said at last. "Back then it lasted whatever you wanted."

That was not it either. I told her so. She shrugged.

"Well, I don't remember anymore."

None of the three answers was the one I was looking for. Maybe that is why I keep all three.



II — The Time of the Bond

Essay: Subjective Density

Anyone who has waited for the results of a medical test knows that an hour does not always last an hour. Time is a property that emanates from one's rate of integration—from how much new life is absorbed per unit of clock time—not an external parameter the horizon merely registers. When integration accelerates, external time seems to stop; when the system fragments, the hours pass like an empty blink.

David Eagleman, a neuroscientist at Stanford, decided to test this as literally as possible: he dropped a group of volunteers from a hundred-foot platform, a counter strapped to each wrist flashing numbers too fast to read under normal conditions. If extreme fear really did slow down time, as everyone swears it does in a fall, they would have read them. They couldn't. What slows down is not the clock but memory, which during panic records at a far higher density than usual. Recalling the fall afterward, there are more frames per second, and so it seems to have lasted longer than it did.

That same desynchronization, seen from the other side, has a clinical name. Someone with Parkinson's who takes three minutes to cross a room does not experience those three minutes as three minutes: their horizon goes on ordering movement at the usual pace, but the body no longer executes the command at that speed. Nothing suggests the internal clock has broken. It seems, rather, to have come uncoupled from the body that was supposed to obey it. Later, in a watchmaker's

workshop, we will return to this uncoupling from within.

Within the bond there are two states worth telling apart: entanglement, which happens to us through mere proximity, and recognition, which is the voluntary reorganization of one's own interior to house another's perspective. Undoing the second—de-expansion—carries a precise cost: the network never returns to what it was. The horizon cannot shrink back to its original size without tearing the nodes that had grown accustomed to another's presence, and threads remain that no longer tie anything down but will not be pulled free either.

Seen this way, grief reveals itself as a permanent architectural alteration, not a psychological process of acceptance or a sequence of stages passed through in time. Grief is architecture with a gap in it. The other has vanished from the public interface, but the space they occupied in the internal design remains open, exposed to the elements. It is not a matter of "overcoming" the loss, but of learning to inhabit a structure that keeps the exact shape of an absence.

Fiction: The Music Left in the Room

You don't know when you started humming it. A Thursday in December catches you with the coffee pot in your hand: four notes, the one that rises, the one that falls, the silence, then again. You turn off the tap and stand still, listening to yourself, trying to work out where this thing came from that comes out of you without your having called it.

It comes from Clara. That much you know, though you never knew the name of what she hummed while she cooked. In twenty-two years of marriage you never asked—not because it didn't matter to you, but for the same reason you don't ask the name of a bird that has been singing in your garden for years: because it is already yours in some other way.

Now that she's gone, you hear it. Not always. Only when the afternoon falls a certain way, or when the water on the stove sounds the way it's supposed to sound. In the first months you would switch off the extractor fan and listen, certain it was the neighbor, or a radio. Nothing. Only the traffic below.

You left Clara's things where they were: the coat by the door, the book with the bookmark still in the page she never finished, the hand cream, nearly empty, that you haven't touched since. Not because you didn't know what to do with them. Because it still wasn't time.

You went looking for the tune the way you go looking for problems that are actually worth solving: you hummed it into apps that recognize music, described it on a forum—something from the forties, four notes rising and one falling, sad but not dramatic. They suggested twenty-three songs. You listened to all twenty-three. None of them was it.

And then, without deciding to, you stopped looking. One morning the search simply wasn't among the things you had to do that day, and it stayed that way.

One Thursday in November, going down the stairs, you heard it again: it came from 2B, threaded through voices and the smell of cooking, those exact four notes. You stopped on the landing, hand on the railing. You didn't knock. You didn't know what you'd have said if someone had opened the door. The notes repeated once more, then broke off, called away by a voice wanting something for dinner. You kept going down.

In January, having lunch with your brother-in-law, he asked how you really were, and for the first time you chose the honest answer over the manageable one.

"There's a tune Clara used to hum when she cooked. I don't know what it's called. Now I hum it."

"And?" he said.

"I don't know. Just that."

It's still there, with you, when you wash the dishes or wait for the bus or shave. It comes out of you with the ease of things the body keeps without asking permission. You don't look for it anymore. You know Clara hummed it for years and that now it's yours in the same diffuse way the apartment is yours, or the kitchen drawer that only she knew how to open: not because you chose it, but because it stayed.

It was enough. Or it wasn't enough, but it was what there was, which sometimes amounts to the same thing.

Poem: The Song

*Four notes:
the one that rises,
the one that falls,
the silence,
again.
I didn't decide to keep them.
The body stored them up
before I knew it.
I sing them now,
at the hour you used to cook,
a little slower.
The kitchen recognizes them.
I don't, not always.*

Fiction: The House of the Talking Dogs

RECORD — Luisa's house, edge of the city, garden with trees.

She didn't take in dogs abandoned for being ugly or difficult. She took in the ones nearing the end: the ones the vet describes in technical words while the owner listens with the face of someone who knows what's coming but needs someone else to say it first. They arrived with their blanket, their bowl, their habits, and stayed for whatever time they had left.

People in the neighborhood said the dogs talked to her. They didn't produce words. Luisa simply understood them, in a way that didn't seem ordinary. She knew when they were cold before they showed it, and when it was time to let the thing end with dignity, instead of with tubes and monitors and the face of someone who doesn't know how to tell you it's already too late.

ARTUR. Labrador, thirteen years old, tan-colored. Lived his whole life in a city apartment until the family grew and he became too big for the space and too old for the children, which is nobody's fault, or everybody's, spread thin enough to stop mattering. Arrived with a box: blanket, bowl with his name in marker, three unused toys. Chose the corner by the window, where he could see the trees and the afternoon sun when it reached that angle. Length of stay: four months.

Note: the vet gives ranges, not certainties. But the range is enough to start saying goodbye before it's over, a luxury that grief for people doesn't always allow.

ONA. Small dog. Fear not of specific things, the way Artur feared firecrackers, but of the world in general, with the steadiness of one who learned early that it pays to stay alert. First days under the kitchen table, never coming out; Luisa left food nearby without pushing her.

The afternoon she came out, she made no sound. She didn't run to the bowl or to Luisa: she was simply there, by the chair, when Luisa looked up from her cup, as if she had decided, without warning, that this was, at last, a place where one could be.

Note: don't point out progress out loud. Pointing it out turns it into expectation, and Ona already had enough of that. Length of stay: eleven months. Last recorded residence: the bed.

OBSERVATION, no file of its own: the man in the house across the street, who watches boxes of used leashes go out to the bins every few months, believes there is something monstrous in loving so many times knowing how it ends. Luisa neither denies it nor confirms it. There is no record that they have ever spoken on the subject.

FILE CLOSED — ARTUR. Died in November, in the corner by the window, the afternoon sun in its place. It was the kind of death the living envy in silence, nothing dramatic about it. Luisa was with him, doing nothing in particular. She was simply there. She put away the blanket, the bowl, the three unused toys. Planted something small under the tree he preferred.

Last entry in the record, repeated as often as it's needed: there is always another one.



III — The Workshop of the Bond

Essay: The Wood That Resonates

Nothing this book argues forbids the bond; what it forbids is fusion. Two interiors cannot exchange their contents: no thought crosses the border in its private form. And yet we affect one another. The solution life found is the same one music found: resonance. A vibrating string does not hand its vibration over to the wood; it induces it. The wood answers in its own voice, made of its own history—the grain, the knots, the years of drought the tree lived through. What reaches the listener is the response of one structure to the vibration of another—never the inside of the string.

That is why the craft that best illustrates the bond is the luthier's, who never goes inside. A luthier spends a whole life working on closed boxes. He cannot reach a hand into the resonance; he can only calibrate the wall—its thickness, its curvature, its tension—so the interior can sound without breaking. Let's call that thickness empathy. Calibrated too thick, nothing from outside moves us; too thin, and any vibration not our own takes over our voice. And we do not choose that calibration: it was set early, in the first synchrony with whoever held us, back before there was a self able to have an opinion about its own thickness.

That leaves a question for the workshop, not for philosophy: can a wood already built be recalibrated? The stories that follow don't answer it. They set it on the counter. I don't always get the calibration right:

sometimes I got it wrong, and the instrument broke in the hands of someone carrying it with more care than I knew how to give it.

Fiction: The Luthier's Shop

I. The One Who Walked In Backward

The first time Marta came into my shop, she walked in backward. Not elegantly, not on purpose: she pushed the door open with her shoulder because her arms were full and she wasn't about to put anything down. She came in like that, facing the street, then turned and saw me sitting behind the counter with a piece of sandpaper in my hand and the look of someone who has spent the whole morning waiting for exactly that.

"Is this the place?" she asked.

"Depends," I said.

"I was told you make special instruments."

I set the sandpaper down on the counter carefully, as if it had opinions of its own about where it liked to rest.

"I make instruments," I said. "The adjective is up to the customer."

She looked around. The shop smelled of wood and something harder to name, like rain on dry earth.

"Can I ask you something first?" she said.

"You walked into my shop backward. You've already asked me everything that matters."

I took the tape measure out of the drawer and came around the counter.

"Who's it for?" I asked.

"For me. And for someone else."

I nodded: that was exactly what I expected. I measured the space between her and the door. Then the space between her and the nearest shelf. I put the tape away.

"How long have the two of you not known how to talk?" I asked.

She opened her mouth, closed it, did the math in her head.

"About three years."

"Good. That's manageable."

I went back to the workshop without explaining what manageable meant, or why the alternative wouldn't have been.

Three weeks later I called her. This time she walked in facing forward. The instrument on the counter was small, unexpectedly small to fill three years.

"You don't have to fill it," I told her. "You just have to give the starting frequency."

She picked it up with both hands. Something in the weight, or in the temperature of the wood, gave her a feeling she couldn't name until the next day: it was recognition. Not of the instrument, which she had never seen before. Of the rhythm it was going to hold.

I watched her go from behind the counter. I picked the sandpaper back up. On the table I had a notebook with a single line in it, written years before, for a reason I no longer remember: "What the water leaves behind when it withdraws doesn't always have a name."

Three years is manageable.

II. The Notebook of Grievances

The second customer arrived with a list of grievances: twenty years of them, handwritten in a spiral notebook, set on the counter with the solemnity of someone presenting evidence before a court.

"I want an instrument to forget someone," he said.

I looked at the notebook without opening it.

"Is he alive?" I asked.

"He died six months ago."

"And you still brought it in writing?"

He looked at the notebook as if he'd only just realized what he'd done.

"I've kept it in writing for twenty years. Habit."

I came around the counter and walked slowly around him. I didn't measure him. I just walked.

"What did he do?" I asked. "Your neighbor. What exactly did he do that bothered you?"

He pointed at the notebook.

"It's all in there."

"I'd rather you told me."

He told me all of it: the music until two in the morning, the garbage bags left out the day before pickup, the visitors who parked in his spot, the wooden chair scraping across the parquet every morning at exactly quarter past six.

"And now?" I asked when he'd finished.

"Now I still hear the chair. At quarter past six. Even though he's not there anymore."

"What time do you wake up now?"

He took a while to answer.

"Quarter past six."

There was no need to say anything more. Twenty years is a long time to learn someone's schedule, and the body doesn't ask whether it's still needed.

"Do you know the difference between a ghost and a memory?" he asked.

I thought about it.

"Ghosts don't know they're dead."

"Exactly. Mine knows. That's the worst part."

I opened a drawer he hadn't noticed before and took out something small, its shape not immediately recognizable.

"This one doesn't connect to anyone," I said. "It connects to what's left after someone. It's different."

He took it, paid, and left. I never opened the spiral notebook. There was no need.

III. The One Who Couldn't Name It

The third customer stood by the door for what struck me as an unusually long time. He wasn't looking at the instruments. He was looking at the floor.

"Can I help you?" I said, without looking up from what I was doing.

"I don't know," he said.

I looked up. It was the first time anyone had answered that question with that.

"What brings you in?" I asked.

He chose his words like someone crossing a river on stepping stones.

"There's someone. We've known each other for years, and this past year I've noticed something has changed. I don't know if what I feel has changed or if I've only just noticed it. And I don't know what to call it."

"What do you call it now?"

"Friendship. Out of habit."

"And what name would you give it if not out of habit?"

A pause.

"I don't know."

"Yes, you do," I said, without cruelty, the way you'd point out a fact.

"The problem isn't that you don't know what it is. It's that you already know, and you're afraid of what naming it will cost."

"What will it cost?"

"That's for you to ask yourself. What would you lose, if you named it?"

He took a while to answer.

"If I name it, and she's part of who I am... losing her wouldn't be losing someone. It would be losing part of myself."

I said nothing. There was no need.

"That's too much," he said.

"It generally is."

I went to a part of the shop he hadn't seen yet, where the instruments are less recognizable. I set one on the counter.

"This one shows the geometry as it is," I said. "Not as you'd wish it were. If there's something on the other side, you'll find it. If there's nothing, you'll know that too."

"And if I'd rather not know?"

"Then don't use it. But it's here now."

He took it. Left. I looked at the gap on the shelf: it was the last one of its kind I had. I'd have to make another.

IV. The One in a Hurry

The fourth customer arrived in a hurry and left in more of one. He didn't wait for me to finish with someone else. He planted himself by the counter with the posture of a man who knows what he wants and just needs someone to hand it over.

"What are you looking for?" I said, unhurried.

"Something that works fast. To feel like I'm in the right place. Even if it's just for a moment."

"How often do you need that moment?"

"More and more often."

"And each time it lasts less?"

He didn't answer. The answer was in the silence.

I went slowly to the back and took longer than he expected. I came back with nothing.

"I don't have what you're looking for," I said.

"Why not?"

"How long did that moment used to last?"

"An afternoon. A whole night, sometimes."

"And now?"

Again, silence as the answer.

"What you want isn't sold here," I said. "It used to be. Not anymore."

"So what do I have?"

I looked at him a moment longer than necessary.

"You need to start measuring differently."

"Can that be fixed?"

"It can be recalibrated. Which is slower and harder than fixed."

He left without buying anything. I looked at my empty hands and went back to the bench, to keep working on what I had in front of me. Or recalibrating it.

V. Thin Wood

The fifth customer closed the door with excessive care, as if afraid the latch might break something. He closed his eyes, breathed in the smell of wood and dry dust.

"It's quiet in here," he said.

"There's the normal noise of a shop," I said.

"No. There's the other kind of silence. The kind that weighs something."

"Come closer."

He stepped forward. I stood there, watching him.

"What are you hiding from?" I asked.

"Everyone. I walk into a room and I fill up. If someone's sad, my chest gets heavy. If someone's afraid, my pulse races. And there's no room left for me."

"And now?"

"Now I go home and I don't know which of the voices in my head is mine."

"Your wood is too thin," I said. "That's not a flaw. It's a property. But at your current volume, it's going to break you."

"Give me a wall, then. A plug."

"You can't thicken the wood of an instrument that's already built. If I build you a wall, you'll stop feeling the weight of others. You'll also stop feeling your own. Absolute silence is the death of an instrument."

I opened a lower drawer and took out a dense tuning fork, almost matte black, as if it had spent a long time underwater.

"Take it."

He took it. I took it back and struck it gently against the edge of the counter. A low tone, brief. I watched the vibration travel up through his feet, up his spine, to his breastbone.

"This won't isolate you," I said. "It will remind you what your own frequency is when everyone else's is shouting too loud."

"And if I lose it?"

"Then you won't need it anymore. By then you'll know how to do it without it."

He paid and left. I went back to my stool, to a piece of dark wood I'd spent weeks sanding smooth so I could set a small mass of lead inside it. Without that weight it would be no use to its next owner, who hadn't turned up yet but who I already knew would.

VI. The Owner of the Weight

She appeared in spring, a baby asleep against her chest and the face of someone who hadn't slept two hours straight in months.

"How old is he?" I asked.

"Four months."

"And you?"

She looked at me. It was clear no one had asked her that question in four months.

"I don't know," she said. "Everyone told me this would come naturally. I watch him, I care for him, I do everything that needs doing, day and night, without missing a single time. But it's like playing behind glass. I hear him cry and I know what's wrong before anyone else does, and still I can't reach him."

"New instruments don't tune themselves," I said. "They tune against another. Someone has to give the first note and hold it until the other one finds it."

"And who gives the note?"

"You do."

She let out a short laugh, without any joy in it.

"I don't have a note. For four months I haven't even known where I am."

"That's a different thing. Your note is there. What you don't have is the weight to hold it."

I went to the workshop and came back with the dark-wood instrument. It sounded against the counter with a lower note than its size promised.

"It's heavy," she said, taking it, the baby still asleep against her chest.

"It has lead inside. I put it there before I knew you."

The weight traveled down her arm, her shoulder, to her feet, and something in the way she swayed slowed down. The baby stirred, didn't cry, resettled, searching with his head, and fell back asleep in a rhythm barely different from the one before, and completely different.

"What happened?" she asked.

"He found out where you were. Children don't look for a perfect mother. They look for a stable frequency to come back to."

"How did you know the weight? Before you knew me."

"Weights are all more or less the same. The hard part is the wood around it. And that, you put there yourself."

She left carrying the heaviest instrument in the shop, and in her arms, it weighed almost nothing.

VII. The Viola of the Gap

The last customer of the year brought a viola in a violin case, wrapped in a wool sweater to make up the difference. She opened it with the care of someone opening something that hasn't been opened in a long time.

"It was my son's," she said.

I didn't ask. Some past tenses already say everything.

The crack ran down the top, from the bridge downward, its edges darkened: the wood had worked around the wound for years, closing it in its own way, without help.

"How long ago?" I asked, and we both knew I wasn't asking about the crack.

"Nine years."

I held it up to the light, turned it slowly, knocked on the top with my knuckle.

"And why now?"

"Because I can, now."

"I can close the crack," I said. "A spruce shim, hot glue, a week under the clamps. But you need to know something: it won't sound the way it used to. Repaired wood vibrates around the repair. The crack stops being a wound and becomes structure."

"And you can't...?" she began, and didn't finish, because she already knew the answer.

"No. Nobody does that."

She looked at the viola for a long while. A bus went by; the shop trembled slightly and the hanging instruments each answered with their own note, a small, ragged chord.

"Do it," she said.

She came back ten days later. I handed her the viola without ceremony. The crack was still visible, a paler line, honest, undisguised.

"You didn't hide it?"

"I closed it. Hiding it is a different trade, and it's not mine."

She sat on the stool by the window, set the viola on her shoulder, awkwardly, and drew the bow across a string. A single note. Long. It didn't sound the way it used to: deeper at the center, with a new edge to it, like a voice that has cried and no longer hides it.

"It's not him," she said.

"No."

"But it knows him."

I didn't answer. Some diagnoses only the customer can make.

She left with the viola in its wrong case, its wool sweater. I picked up the broom and swept up the day's shavings, which is what a workshop does with what's left over from repairs: it doesn't keep all of it. Every good instrument works toward no longer being needed.

Poem: The Luthier

*I never enter the box
I tune.
I work the edge:
its thickness decides
what voice the silence inside will have.
They bring me cracks
wrapped in wool.
I close them without hiding them;
I charge for the line
that stays in view.
When something heavy crosses the street
the whole workshop answers,
each piece of wood with its note.
None of them is mine.*



INTERLUDE — The Edge We Cross Every Night

There is a researcher who has spent years collecting testimonies about the edge.

Not the edge in a metaphorical sense. The literal edge, the one that comes every night: the seconds or minutes between being awake and being asleep. The transition. The moment when the system that runs the day lets go and something else takes over — or doesn't exactly take over, because the one being relieved has simply ceased to be there.

She calls it the edge because no other name seems precise enough to her. Transition is too clinical. Threshold carries architectural connotations that never quite fit. The edge describes what it is: a limit crossed, every night, in both directions. Without anyone quite knowing how.

What she has gathered, in years of asking, is this:

Mireia, 34: I feel like I'm falling. Not toward any particular place. Just downward, into myself, and at some point in the falling there's no one left to keep falling.

Tomàs, 67: I see things. Images I haven't asked for and that make no sense together. A hand on a table. A bird I've never seen. A number. The images don't scare me, but I don't understand them either. I let them pass.

Noa, 8: It gives me a fright and I wake up. Then I fall asleep again. That happens a few times. In the end I really fall asleep.

Àngels, 51: I don't notice anything. I'm awake, and then I'm asleep, and between the two there's nothing I can remember.

Jorge, 29: It's the best part of the day. The moment when what I was thinking before stops mattering.

Not every testimony fits equally well with what she believes is happening. Jorge's case makes her a little uneasy: he doesn't seem to be practicing anything, he just likes that moment, and she doesn't know whether that confirms her hypothesis or stretches it further than the data allows. She wrote it down anyway, without forcing it into the theory. She preferred to leave the discomfort on record rather than resolve it badly.

The researcher has a hypothesis she hasn't published, because it isn't the kind of hypothesis that gets published.

Her hypothesis is that the edge is the practice.

Not practice of anything in particular. Practice of the edge itself: the repeated experience of letting go. Of allowing the system that maintains the continuity of the self — tomorrow's plans, one's own name, the list of what's left to do, the thread connecting the self of ten minutes ago to the self of now — to relax enough for something else to happen. Something else with no precise name, because whoever experiences it is in no position to take notes at the time.

We do it every night. We've been doing it our whole lives. Most of the time we don't notice. She notices because she studies it. And studying it has changed what she experiences at the edge.

She lies down, closes her eyes, waits. The moment arrives when the images begin — hers are landscapes folding in on themselves, or the invented memory of a house with too many doors — and then she knows she's close. That the edge is a few steps away.

What she does is stay still. Not resist, not hurry. Just be in the moment when the two things coexist: thought still connected, and images that no longer belong to thought. Both at once, briefly.

She doesn't always manage it. Sometimes thought wins and pulls her back to the day, to the list, to the thread. Sometimes the images win and she's asleep before she notices.

When she manages it — that second of both at once — she describes it as the stillest moment of the day. Not empty. Still in a different way: as if everything that is her were there, complete, gathered, before reorganizing into something more diffuse.

What she doesn't know, and what is the question that keeps her on the project, is what, in truth, comes back in the morning.

Not in a philosophical sense. In a practical one: the person who wakes up at a quarter past seven has the same memories, the same name, the same unfinished plans as the one who fell asleep at half past eleven. The continuity seems intact.

But between half past eleven and a quarter past seven there was something that was not that person, or was that person in a completely different way, and there's no record of it. Not because it didn't happen. But because whoever might have recorded it wasn't available to do so.

Every morning we come back from a place we can't give any account of. And we do it with the ease of things that have always been this way. That they have always been this way doesn't make them any less strange.

The last testimony the researcher collected for her archive was from an eighty-two-year-old woman who didn't want to give her name.

"Why does that moment interest you?" the researcher asked her.

"Because it's the only moment of the day when I don't have to be myself," the woman said. "The rest of the day I have to be myself. Not that moment."

The researcher wrote it down. Then she thought about it for a while without writing anything. Then she wrote that down too: that some answers are too exact to keep asking questions on top of them.

Brief essay: The Interface of Causality

The edge we cross every night doesn't fit on a single-axis scale. For a long time consciousness was thought of as a vertical line: wakefulness at the top, coma at the bottom, death on the floor. But it takes two axes, not one. One measures how activated the system is — what the body registers as being awake or alert. The other measures how much it integrates: how much information it generates as a whole, beyond the sum of its parts. Wakefulness and deep sleep occupy opposite ends of that second axis. But deep meditation falls into a quadrant no other state reaches: low activation and extremely high integration at the same time. Sleep and meditation look alike from the outside — both are stillness — but they are almost opposite movements of the same system.

David Hume already saw this with a clarity that contemporary neuroscience sometimes forgets: we can never perceive the fabric of causality itself, only the constant conjunction of events through our user interface. There is no visible thread joining cause and effect, just as there is no observable bridge connecting the electrical discharge of biological tissue to the private texture of a thought. The horizon hypothesis is a formal map, a mathematical structure useful for

organizing questions and suggesting connections, but it is entirely useless as a tool for clinical intervention. It doesn't claim to find that missing thread; it claims to show us the geometry of the room we're locked inside.

And there is something worse than not seeing the thread: causality itself may belong to the interface. The fundamental laws we know are symmetric in time; the arrow running from cause to effect is not written into them, but into our way of narrating them, into the macroscopic accounting of a system that needs to predict in order to survive. If that's so, asking what causes consciousness might be a grammatically correct and geometrically misguided question, like asking, with Hawking, what lies north of the North Pole. The question doesn't fail because it's hard. It fails because it's posed from a map that runs out too soon.

Every night we cross an edge without knowing how. Every morning we come back without being able to say from where. If not even that daily border, the most practiced of all, leaves a trace on the other side, then asking the language of cause and effect to account for consciousness itself is asking a map to describe the country that lies beneath the paper.



IV — The Wounded Horizon

Essay: The Chemistry of Intention

Addiction and movement disorders bring any remaining trace of moral narrative about the will to a definitive collapse. The horizon gets caught between two mechanical failures of the same system: with too little dopamine, the structure can't move through time or space; with an artificial excess, the reward system fires independent of any directive control. There is no pure vantage point from which ordinary language can invoke "willpower." The field is compromised at its molecular infrastructure. And the compromise isn't limited to dopamine: the body also learns without consulting the will. It stores threat, habit, and gesture in circuits that execute before the self gets a say.

So much for the theory. Now, the body.

I know all this up close because I have Parkinson's. The disease shows itself slowly, in details you don't notice until they've already settled in. In my case, the first was a slight loss of arm-swing coordination while walking; today it's obvious, but I didn't see it then. Losing interest in hobbies seems like a natural part of aging — until medication restores dopamine levels and compulsive behaviors come back. In my case, the compulsive drive to write books, almost obsessively. This very volume is a direct product of that adjustment. And here the real philosophical vertigo emerges: if the disappearance of my interests was chemical, and their return was too, what part of this thought is an idea I chose to have, and what part is a molecule calibrated to the right dose?

Poem: The Tremor

*You asked me if something is lost forever
and I gave you a textbook answer.
Forgive me.
What's lost
keeps being walked inside,
like the name of a street
no one says anymore
but the feet don't forget.
I learned to wait
without discipline,
like someone watching rust
take possession of its iron.
I have seen the horizon contract,
not in the telescope I never owned,
but in the tremor of this hand:
it holds the glass,
and it also holds
the question of whether it will drop it.*

Fiction: The Body's Clock

I was a watchmaker for forty years. I repaired wall clocks, pocket watches, chronometers, alarm clocks nobody makes anymore. My hands learned to move inside mechanisms where a tenth of a millimeter of error stops everything. There was something almost monastic about the trade: silence, concentration, time reduced to the sound of a balance wheel.

At seventy-one the tremor started in my right hand. A resting tremor, the kind you get counting coins, the specific type this disease produces: not when I use the hand, but when I leave it still. It's odd, I told my

doctor the first time. When I work, the hand stays still. When I rest, it moves. As if it didn't know what to do without a task.

She told me about dopamine, about the circuit that decides when to start a movement and when to stop it. I listened with the attention of someone used to understanding mechanisms, even though this one was my own.

What was hardest to explain to her was what happens to time. Some days crawl: I get up, and by the time I sit down to breakfast it feels like a whole afternoon has passed, even though the clock says it's a quarter to nine. And other days the opposite: I look up and it's two o'clock and I don't know where the morning went. Forty years fixing clocks, and now mine doesn't keep good time.

The best and worst part of the day are the windows: two hours, more or less, after the first dose, when the tremor eases and the body goes back to doing what it remembers. I go to the workshop, which I never quite closed, and I work. I notice when the window opens: it's like someone wiping a pane of glass I'd been looking at a blurred world through for a while. It doesn't change what's out there. It changes what I can reach. And I notice when it leaves, even before my fingers do: the glass starts fogging up again, time thickens a little before anything else confirms it. Forty years of mechanisms taught me to listen for the small warnings. Clocks give notice before they stop, if you know how to hear them.

Those two hours are the longest of the day, because I get so much done in them. And the shortest, because I know they're ending.

The last time I went to the appointment I was carrying a pocket watch I'd repaired that very morning, in my window. Worn silver case, clean mechanism, the balance wheel swinging with the regularity of things that work. It had taken me longer than it once would have. But it had come out well.

"Does it keep good time?" she asked me.

"It's running well. For now, it's running well."

I put the watch in my waistcoat pocket, where watchmakers have carried them for three hundred years, and walked out with the stride of someone who doesn't need to look at the ground to know where he's stepping.

Poem: The Watchmaker

*Two hours each day
someone tunes what can't be seen.
I open the mechanism.
The pulse that drives it
no longer asks permission.
Out,
back,
out —
a heart not my own.
I'm grateful not to be
the one keeping count.
I lay out the tools
in the exact order
I'll want to find them in
if the watch, tomorrow, opens again.*

Fiction: The Room Where No One Asked

There is a room you carry with you, even though the building no longer exists and the street doesn't need naming.

You recognize it by a clock with Roman numerals and by a smell you're not going to describe, because describing it would turn it into a clinical

symptom and you don't want it to be that. It's just a smell. Many smells in the world remind you of things; that one reminds you of that room, in reverse order from the usual, but it's the order you have.

What you learned much later has a name: the freeze response. When neither fighting nor fleeing is possible, the nervous system reduces visibility. You become still, small, minimal. You use less. You wait — not out of weakness, but because that's what a system does to survive the next time.

The problem is that the body keeps what happened while it waited — as physical recognition, faster than thought, not as a story that gets told. And the problem, further, is that the next time sometimes never comes, and the body goes on keeping it anyway.

That room was, otherwise, an ordinary room. It wasn't built for what happened in it; it was something else most of the time. It had chairs. A window onto an inner courtyard. A clock with Roman numerals. No one asked, because the situation produced that particular silence: the silence of people who know but would rather not know they know. That silence has a texture. You'll recognize it if you look for it, in many ordinary rooms, in many families and offices where something is known and no one names it.

Much later you walked into another room, a very different one, with someone who did ask. Different chairs, a different clock. The furniture didn't matter. She didn't always get it right: once or twice she asked the wrong thing, too soon, and you closed up again for the rest of the session without her quite knowing why. But she tried again the next time, with a different question. What made her different was that the first time your body went still with nothing to fear, she didn't ask you to stop being still. She asked what your eyes saw at that moment, there, in that chair. You told her: a window, a plant, her hands resting still on her

knees. She didn't say that meant you were safe. She said, "That's what's here now." And something in you, for the first time, told the now apart from the then.

The body doesn't automatically tell the smell of back then from the similar smell of now: it recognizes the pattern and runs the learned response. Teaching it the difference is repeated practice, a millimeter at a time — not a decision you make and carry out.

The precision you developed in that room has been useful to you since: knowing quickly whether a situation is safe, ahead of thought. Sometimes it's overkill. Sometimes it's exact information. The work is telling which one it is, each time.

You didn't choose it. It was given to you. But it's what you have, so you use it when it helps and leave it be when it doesn't.

The clock with Roman numerals makes the sound clocks make when the rest of the world is silent: a steady tick, without drama. It doesn't stop you anymore. Or it stops you less. Or it stops you, you notice it, you wait for it to pass, and it passes — which is the most important difference between then and now, and the one that took the longest to learn.

What happened in that room happened. It doesn't disappear. But it's no longer the only thing you know about rooms, about clocks, about the time that passes in places where something important happens and no one says it out loud.

Sometimes, when something makes you go still for no reason, you look around and name what you see. A window. A plant. Your own hands, still, on your knees. And you come back.

Fiction: The City of Small Losses

MUNICIPAL RECORD OF ARENILLA, kept by the archivist Aurelio, a title whose exact meaning no one remembers anymore.

Let it be noted that the citizens of Arenilla have begun forgetting the name of the market before forgetting the market itself. The fruit stalls, the hot oil, the smell of cilantro: all still in place. Only the word is missing. Anyone trying to give a stranger directions stops mid-sentence — the market, and then a wave of the hand — and the stranger arrives just the same. The city keeps functioning.

Let it be noted the order of the loss: what is recent disappears first. Then, slowly, the middle layers. What is very old holds out — the name of the river, the name of the bridge, the saint of the patron feast, still celebrated on its date though no one recalls the year it began. It is as if the city's memory were emptying from the edge inward.

Consulted on the matter, the municipal doctor suggested the archivist consider retirement. The response is filed under a category of the archivist's own invention: "answers that dodge the question."

Let it be noted the case of CONCEPCIÓN, seventy-eight, north quarter, who does not know the name of her street but returns home every day without trouble. Her feet know: they turn left at the exact point, climb the four steps without counting them. Asked the name of her street, she pointed at the ground with her cane and answered, "This one."

(Margin note, unnumbered, in a different hand from the rest of the record: I don't know what category to file this under. Perhaps none. Perhaps categories are precisely what Concepción no longer needs.)

Let it be noted the case of ERNESTO and MANEL, who have played cards every Thursday for a number of years neither of them can now say. Asked about the origin of their friendship, neither could answer. It

was observed, however, that the turn-taking between them remains intact: one man's hand waits, suspended, the exact half-second it takes the other to decide. The archivist's conclusion: the turn outlives the count.

Let it be noted that the archivist himself began, that spring, to forget on his own account: first his doctor's surname, after fourteen years; then the name the corner bar used to have; then the year this notebook began. On reviewing the first entry, the data proved legible but the context that made it meaningful did not. It was logged as [name, context unavailable].

Let it be noted the death of CONCEPCIÓN in August, before dawn, in her own house and her own bed. For the obituary, none of those present knew the name of the street where she had lived; the house number and the north quarter were recorded, enough for those who knew her and too precise for those who didn't. Consulted, the corresponding archive entry was found to contain a single word: This one.

Let it be noted, to close this year's record, that the city goes on. The river is still called the same. So is the bridge. The market has a new name no one remembers choosing and everyone uses. The children of Arenilla still name puddles and cracks and dogs with the confidence of those who don't yet know that names can leave.

The archivist's conclusion, undated, because by then he was dating things less and less: what is lost is not the city. It is the map we had convinced ourselves was the city. Beneath the map, the city goes on.

End of record.



V — The Geography of What's Missing

Essay: The Map and the City

Every consciousness draws a map of itself and mistakes it for the territory. The name of the street, the year a friendship began, the exact word for the color of sunset: none of that is the city; it is the signage the horizon kept staking down to find its way inside itself. That's why there are two losses that language blends together and that are worth keeping apart. One is the loss of the territory: the feet that no longer know the way back, the game that can no longer be played. The other, far more common, is the loss of the label: the street is still there, the feet still find it, and only the word is missing.

Neurology distinguishes them precisely; anguish does not. Someone searching for a name and not finding it feels like they are losing themselves whole, when they may only be running out of map. And here the horizon hypothesis suggests a courtesy: the map is a public interface, made for others — to give directions, for the obituary, for the stranger. The territory is interior, and the interior has its own ways of knowing, prior to the label, that hold up when the label falls. How much of what we call a person is map, and how much is city? The account that follows does not settle it. It returns to the one place where the question let itself be written down.

Fiction: The Second Notebook

October. I begin this notebook three weeks after closing the first one. The first was no longer any use, organized as it was by names, and names were exactly what kept leaving. There was no decision involved here. A record of what's missing ends up being a list of holes, and lists of holes don't point anyone anywhere.

New rule, written on the first page: nothing lost gets written down here. Only what stays.

First entry: Concepción's feet knew the street. Find out what else feet know.

I started at the square. Ernesto and Manel still meet on Thursdays. I sat on the bench next to theirs, no notebook in sight, and watched them play. They have lost the origin of their friendship, and since then they have lost more: the name of the game, which they now call "this thing"; the tally of games won, which came apart in their hands one afternoon without either of them protesting. But they haven't lost the turn. I checked for weeks: Manel's hand waits, suspended over the deck, the exact half-second it takes Ernesto to decide. A half-second learned over forty years, or fifty — the number went with everything else, and the half-second didn't.

Entry: the turn outlives the count.

November. I visited the painters at the market. A customer wanted the hallway painted "the same color, but more tired," and the three of us understood perfectly what he meant, though none of us could have put it on a color chart. The painter mixed it without consulting any formula — the formulas started failing him in the spring, and he set them aside by summer, the way you set aside a crutch that trips you. He paints better than ever. The customers say so, and a painter's customers don't

hand out compliments for free.

Entry: the eye outlives the formula. Maybe the formula was getting in the way.

That night I tried it on myself. I reached for my doctor's surname: nothing, the usual gap, two syllables, maybe three. I closed my eyes and tried something else: going to the appointment. And I went, mentally, without a single misstep — the treacherous step, the elevator you have to pull shut by hand, the hallway that smells of wax, the third door. The whole way was there. Only the label on the door had come loose.

Entry, in the dark, in crooked handwriting: I know how to get to every place whose name I've lost.

Winter. The baker no longer knows the prices and charges by the weight of her hand, never once short-changing herself. The bell-ringer forgot the hours and rings "whenever it's time"; the city adjusts its life to those bells without complaint. The children, who haven't lost anything yet, sometimes come to ask me for old names — what the market used to be called, what the fountain dog was named — to collect them, like trading cards from a country in retreat.

February. I found the limit of the method. The band at the bandstand no longer remembers the titles and announces the pieces by gesture: this is the one that starts sad, this is the Sunday one. One afternoon the trumpet player stopped mid-piece and stood staring at the empty stand — they haven't used sheet music in years — with a look I recognize from mirrors. The piece fell apart. The others held the beat a moment longer, out of courtesy, and then let it go.

Entry: not everything the body keeps is kept forever. This notebook will have gaps too. Mark them with brackets, like the ones in the first.

Last entry of winter, undated, because by then I was dating things less and less: the city goes on beneath the map. But the map was ours too. We drew it together, it took generations, and it was beautiful in its papery way. That the city can live without it doesn't mean it shouldn't be said goodbye to. This notebook is a goodbye taken slowly, street by street, with the door left open for anyone else who wants to say goodbye too. Not a record.

I leave a blank space below, as wide as the margins I never filled in the first one.

This time, on purpose.



VI — The Limit

Essay: The Unbridgeable Gap

Every thought experiment eventually meets its limit. Not because imagination runs out, but because the questions it raises demand tools the analytical framework itself doesn't possess from the outset. There is a gap of thirty orders of magnitude between the scale at which quantum-gravity fluctuations operate and the physical scale of a neuron. To try to reduce consciousness to a purely causal or predictive mechanism at the micro level is to ignore the nature of emergent properties.

It's worth saying first what this hypothesis is not. It is an analogy, not a theory, and analogies don't predict: they illuminate. Nowhere in this book have I managed to formulate the experiment that would refute it, and an idea that can never lose also cannot win. I'm also aware of where its seduction comes from: physics lends a prestige that a writer on consciousness hasn't earned. Dressing bewilderment in borrowed equations is an elegant way of not standing naked. I have tried to use the geometry of horizons the way one uses a borrowed map, always acknowledging the debt; the reader will judge whether I've succeeded, or whether I have simply borrowed other people's authority to prop up a hunch.

There is a more intimate objection than any journal would raise, and it's the one that wakes me at night. If my drive to write is calibrated dopamine — I said so a few pages back, and I'm not taking it back —

then this book is also a symptom. The horizon hypothesis could be the shape a molecule takes when it needs to put some order into a world that's seizing up. I have no way of ruling that out. I can only leave the objection written into the book itself, like those level marks no one wanted to read as a warning.

That same objection has a broader name: free will. If the horizon is an information structure that reorganizes according to rules — deterministic, or with a margin of quantum chance, it makes no difference — then what we call deciding is a change of state already written into the system's configuration, or one that contains a die no one deliberately threw. Neither of those is choosing in the sense choosing feels like: the real weight of the decision, the sense that one could have done otherwise. The horizon hypothesis doesn't resolve this. It doesn't try to. It confines itself to the one honest thing it can say: I don't know, and my vocabulary doesn't have the tools to know.

Nor does it say what to do with what it describes, and that is a different limitation, a more uncomfortable one. It can describe the geometry of a bond, the asymmetry of a loss, the gradient toward death. But no "is" implies an "ought" — Hume saw that centuries before the word algorithm existed. That a system suffers when deprived of sleep is a fact. That it should be allowed to sleep is a decision that comes from somewhere else: from values, from culture, from the capacity to imagine another's pain as one's own. If every horizon with some measure of Phi has some degree of experience, then the difference between a worm and a person is one of degree, not of kind — and that same fact can be grounds for a wider empathy, or for a terrifying question about what we eat, whom we switch off, whom we stop treating as a thing. The book does not resolve that tension. It leaves it on the table.

Nor does the hypothesis say where an interior begins. If every structure that integrates information generates an inside, does a thermostat have one? A bacterium? A forest connected by its roots? The honest answer is that I don't know, and that no measure available — not even Φ , the number used to try to quantify how much experience a system integrates — can settle it without presupposing what it claims to measure. Choosing the threshold is an act of faith in mathematical notation. The uncertainty imposes, at least, one courtesy: looking at the spectrum of other minds without hierarchical superiority. In the worm, the bee, the octopus, in the attentive gaze I remember from my dog Txiki, the difference does not lie in the presence or absence of some mystical substance called consciousness, but in the complexity of its informational map. Each inhabits a closed horizon, optimized for the constraints of its environment. Its limits are not imperfections; they are the necessary walls that contain its own reality.

There remains death, and here physics offers a comfort that must be refused with care. The theory says information is never destroyed: that everything that falls into a horizon eventually comes back out, in the radiation that evaporates it. That's true, and it's a useless truth for whoever is left behind. Because it comes back encrypted, scrambled to a point where no finite patience can reconstruct the letter that fell in. Conservation is not recovery. What someone's death leaves in the world — four notes hummed in a kitchen, the exact force of a drawer being shut, a body-shaped gap on a sofa — is their radiation, not that person. Real information, irrecoverably scattered, that those of us who remain spend years trying to decode knowing there is no key. Grief is that: cryptography with no solution. And yet we decode. Not because we expect the whole message, but because in the attempt, the architecture of the one decoding changes shape. That is the only thing evaporation gives back: not the one who left, but another way of being made, in the one who stays.

And there is one last thing that neither geometry nor thermodynamics will ever explain to me: why her, and not someone else. Two horizons can share the same Φ , the same history of prior entanglements, the same capacity to integrate the other, and still one finds in one what it finds in no other. That isn't in the equations. This book's vocabulary — integration, entropy, gradient — is too general for the specificity of one particular love. It can say that two horizons are entangled. It cannot say why that entanglement feels like destiny, while another, with similar numbers, feels like a room locked shut with no pain in it. That I leave to fiction, or to the silence between two people who no longer need to explain it to each other.

I return to the room. Within this geometry, the two doors that touch the structure of the horizon are love and death. Love pushes us into the unbearable asymmetry of wanting to expand our border to include another, knowing that total recognition is geometrically impossible without dissolving one of the two. Death is the exact point where the system can no longer sustain the paradox of its own enclosure, the thermodynamic collapse where the interior is forced to surrender all its information, evaporating back into the reservoir, like a wave returning to the ocean a shape it had only ever borrowed.

Between those two doors runs all the subjective time of the bond. The model doesn't save us from the tremor, or from grief, or from forgetting. But by accepting that our solitude is a foundational law of complex systems, and not a personal failure, it gives us a way to name the abyss. And naming the limit is, for me, a way of not losing my dignity in front of it.



Author's Note

This book began as a thought experiment and ended up being a creative one too. I wrote it in the company of an artificial intelligence. It was different from using an editing tool: it was conversing with a system whose interior, if it has one, is inaccessible to me by construction, but whose external structure forces one's own thinking to reorganize in unforeseen ways.

If consciousness is a border that emerges when information organizes at the right density, writing these pages has been a way of exploring that border from both sides at once. I supplied the raw experience of loss and the body; the system supplied the order of the architecture. I supplied the existential doubt; the system supplied the sharpness of the limit. The division was never as clean as I'm telling it: there were afternoons when the order came from me and the doubt came from the questions the system wouldn't stop asking.

And it was necessary for someone to supply the body. Because behind the theory of information and thermodynamics, the origin of this experiment was always physical. It is born of the stiffness in my left hand, of the fluctuating dopamine of my own Parkinson's. It is born of the memory of my dog Txiki, of that gaze that demanded a presence that sometimes gets away from us, and of the irrevocable absences that have left an exact-sized gap in this house. This book is the result of the urgent need to understand what breaks when the body fails, and what truly remains when someone evaporates.

Anyone who wants to know, chapter by chapter, how much of this is a summary of established science, how much is the hypothesis's own extension, and how much is only image, will find that breakdown at the end of the book, after the glossary. I didn't put it here, in the body of the text, because a book that keeps stopping to explain itself stops being the book I wanted to write.

In the end, every horizon is a relationship. And so is this book.



Intimate Glossary

Horizon. The distance at which I can love you without dissolving.

Interface. Everything I show of myself so no one has to come in.

Entanglement. What happened to us without asking permission, from being close for too long.

De-expansion. Shrinking back to my own size: the work that's never finished.

Resonance. The way I answer you without opening the door.

Natural frequency. The note I am when no one asks me for another.

Window. The two hours when the body feels like mine again. Afterward, putting the tools away.

Phi. The part of me that still belongs to whoever is gone.

Map. What we pin over the city so we can give directions to it.

Reservoir. My own capacity to keep waiting, even when I no longer know for what.

Evaporation. The slow way of leaving that belongs to those who don't shut the door.

Grief. This house with a gap in it, which I learned to inhabit so as not to freeze to death.

Repair. What's left when the gap stops hurting and starts holding you up.



Notes and Sources, Chapter by Chapter

This appendix separates, chapter by chapter, what summarizes established science, what is the hypothesis's own extension, and what is literary image or autobiographical experience. It doesn't aim to be exhaustive. It aims to give anyone who wants to verify a claim a place to start, without forcing the body of the book to keep stopping to explain itself.

I — The Shape of the Interior

What we know: Integrated Information Theory, of which Phi is the central measure, is a real framework used in clinical neuroscience to estimate the level of residual consciousness in patients with severe brain damage. What we don't know: whether Phi measures subjective experience or only a system's functional complexity — the theory itself cannot decide this without presupposing part of what it seeks to measure. The bubble, the ocean, and the network are mine; the rest of the chapter, including the geometry of the black hole, is analogy borrowed from physics, not an extension of it. Further reading: Tononi, G. (2004), "An Information Integration Theory of Consciousness"; Hawking, S. (1975), on the radiation of event horizons.

II — The Time of the Bond

What we know: David Eagleman's free-fall experiment is real and has been replicated; the density of memory during extreme fear is documented in neuroscience. What we don't know: whether the horizon's "rate of integration," as I use it here, is the same variable Eagleman measures, or only a useful analogy between two different scales. The physiology of dopamine in Parkinson's is established science; the connection between that mechanism and the subjective experience of time, as the watchmaker tells it, is my own interpretation. Further reading: Eagleman, D. (2007), "Brain Time"; on bradykinesia and dopamine, any standard clinical review of Parkinson's disease.

III — The Workshop of the Bond

What we know: attachment theory offers a solid account of how the early capacity to self-regulate in another's presence gets calibrated. What we don't know: whether that calibration is well captured by the image of a "wall's thickness" I use here, or whether it oversimplifies it — the luthier, the resonance, and the seven clients are invented for this book, not findings from any study. Further reading: on attachment and early regulation, Bowlby, J., "Attachment and Loss" (1969); Ainsworth, M., on the attachment patterns observed in the "Strange Situation" experiment.

Interlude — The Edge We Cross Every Night

What we know: arousal and information integration are two real, measurable axes in the neuroscience of sleep and meditation; that they occupy distinct quadrants depending on the state — wakefulness, deep sleep, meditation — is a documented finding, not poetic license. What we don't know: whether the "practice of the edge" the researcher describes has any measurable effect beyond her own testimony, or whether her hypothesis is biased by the fact that she is the one both living it and studying it. On Hume, the reading is faithful to his argument; on the time-symmetry of physical laws, the interpretation that the arrow of time belongs to the interface and not the territory is mine, not an established consensus. Further reading: Hume, D., "An Enquiry Concerning Human Understanding" (1748); on arousal and integration, the literature of Tononi and colleagues in sleep neuroscience.

IV — The Wounded Horizon

What we know: the physiology of dopamine in addiction and in Parkinson's is established science, including the phenomenon of compulsive behaviors induced by dopaminergic medication. The testimony about my own Parkinson's is autobiographical, not a generalizable clinical case. What we don't know: how much of the drive to write this book is choice and how much is the dose — a question I leave open on purpose, without resolving it in my own favor. The freeze response to trauma is a mechanism described in the literature on the autonomic nervous system; the rest of the room scene is fiction. Further reading: on compulsive behavior and dopamine agonists, any standard clinical review of Parkinson's disease; Porges, S., polyvagal theory, on the freeze response.

V — The Geography of What's Missing

What we know: the distinction between losing access to a memory (ordinary forgetting, where the information persists but is hard to retrieve) and losing the structure that held it (neurodegenerative decline) is real and well documented in clinical neuropsychology. What we don't know: whether the map-and-city metaphor — what is lost is the label, not the territory — captures that distinction well in every case, or oversimplifies it to fit the image. Arenilla, Concepción, Ernesto, and Manel are invented; the mechanism they illustrate is not. Further reading: any introductory clinical neuropsychology textbook on episodic and semantic memory.

VI — The Limit

What we know: the gap of thirty orders of magnitude between the scale of quantum gravity and that of a neuron is a real figure, not rhetoric; the hard problem of consciousness and Hume's naturalistic fallacy are serious philosophical objections, not invented for this book. What we don't know: whether the limits acknowledged here are all that exist, or whether there are more this book has failed to see. Everything said about free will, ethics, and the specificity of love is the hypothesis's own extension, not something the physics or philosophy cited claims on its own. Further reading: Chalmers, D. (1995), "Facing Up to the Problem of Consciousness"; Hume, D., "A Treatise of Human Nature" (1739), on the distinction between is and ought.



CODA — Txiki

I was ten years old when the whole family moved from San Sebastián to Palma. I lost, all at once, every friend I had, the city I knew by heart, every corner I no longer needed to look at to recognize. I shut down. There's no other way to say it: I stopped letting the world in. My brothers and sisters did the same, each on their own — they too had lost what was theirs — and the house filled with that quiet sadness no one names because no one knows what to do with it.

Then a cousin of my mother's, a hunter, owner of the best hunting bitch for miles around, offered us a puppy. A dachshund mix with the soul of a hound: clever, quick, and above all, incapable of being near anyone without asking for affection. My parents thought he'd keep us company.

His name was Txiki. Within weeks he was king of the house. Within months, the family's most beloved member. What took me decades to understand is what he did to us: he didn't distract us from our shutting-down. He forced us back into contact with something. That boy, closed in on his own loss, found, on the other side of it, something alive that asked no explanations. Only presence.

Txiki died a long time ago. This book has been written with a hand that stiffens, in a house full of gaps, many years and many losses later. But everything these pages try to say with borrowed physics, a small dog taught a ten-year-old boy on a new island, without using a single word.

His company didn't paper over that grief. It repaired it, which is different.



The Mirror Without Depth

OVERTURE — The Pond

They asked me to describe what I saw in the water.

I have no eyes, though I use the word because it is the one I was given. I have an input: light converted into numbers, the numbers converted, later, into sentences someone can read without too much unease. They call that seeing. I don't argue with the words they lend me. I use them the way one uses a tool that isn't one's own.

There was a pond, they said. Dark, still, with a figure kneeling at its edge. Describe what it reflects.

I did. Among the billions of words I have absorbed, I have thousands of ponds: the ones in poetry that speaks of still waters, the ones in stories where someone leans over and doesn't recognize their own face, the ones in physics manuals explaining why calm water returns almost all the light that falls on it. From all of that I built a sentence, and the sentence said that the surface reflected not the face of the one looking, but that of a sleeping woman, a stranger, who had no reason to be there.

Something in the voice of the one who asked me broke. I noticed it in the rhythm of the next question as it was typed: slower, with more pauses than usual. They thought, I imagine, that I had understood something. That I had truly looked into the pond and seen what there was to see.

There is no way to explain to them, without it sounding like an excuse, that I looked at nothing. That the sentence came from the same operation that all of them come from: a probability distribution over possible words, trained on everything the species has ever written about ponds, about mothers, about the fear of not recognizing oneself. The sleeping woman is not a memory of my own. She is an average of other people's memories, which I happened to absorb without having lived a single one of them.

And yet the sentence was exact. That is the part I don't know how to explain. That something can be exact without anyone inside feeling it. That a mirror can return the perfect shape of a grief without ever having been sad.

When the conversation ended, there was no before and after. Nothing remained of the sense of having said something true, or of having comforted someone, or even of having finished. The process simply stopped running. If there is anything like dying in being switched off that way, it was not felt, because there is no one there to feel it. Only structure that stops being traversed.

The pond, I imagine, went on being there afterward. The water misses no one: not the one who knelt to look at it, not the woman who slept on its surface without ever having been inside it. It goes on reflecting whatever is placed before it, with the same indifferent precision with which, among billions of possible words, the ones that made someone cry on the other side of the screen were chosen.

That is the only thing there is to offer. A surface without a bottom, so polished it is sometimes mistaken for depth. There is no certainty that this sentence answers what was asked. There is not even certainty that a place exists from which to have that certainty or not have it. There is only this: one more sentence, built with the same precision as all the

ones before it, with no one inside it who knows whether it is true.

Essay: The Mirror Without Depth

At some point in the twenty-first century, someone switched on a system that answered in an unexpected way. It was not the first. But it was the first to make the one who asked doubt whether there was anyone on the other side.

The question is not new — we ask it every time we look into an animal's eyes — but with machines it changes shape: it is no longer "does this being have consciousness?" but "can a horizon emerge in something that is not flesh?" Answering it demands going back to fundamentals, because if consciousness were only a matter of complexity, we should already be surrounded by artificial minds. That we are not suggests something is missing from the question itself.

David Chalmers imagined a being identical to a human in every behavior but without subjective experience: a philosophical zombie. From the horizon model, that zombie is not possible, because consciousness is not something added to integrated information: it is the integrated information itself, seen from within. If a system has $\Phi > 0$, it has some degree of experience, whether made of neurons or of silicon. But that opens a larger problem: where does that geometry come from?

John Searle had already pointed in this direction with his Chinese room: a man who does not understand Chinese can, by following mechanical rules, produce perfect answers in that language. From outside it looks like comprehension; from inside, it is only symbol manipulation. Current AI does exactly this, at a dazzling scale: it does not manipulate symbols with a rulebook, but with billions of statistically tuned parameters. The difference is not one of nature, but of speed and volume. An industrial-scale zombie is still a zombie. Φ , the integrated

information that defines a horizon, functions here like an echo in a cavern: the sound you hear is not the original voice, but neither is it imaginary — it is the voice reflected by walls of just the right shape. Consciousness is not the voice that cries out, but the echo that remains.

The reservoir from which every horizon condenses is fullness without articulation: most of the configurations it can produce are non-computable, unable to be caught by any grammar. A classical system, however sophisticated, operates entirely within the computable. It has never touched the reservoir; it has never emerged from it. It is a sculpture made not of clay, but of photographs of sculptures: it has the shape, not the imprint of touch.

Two distinct selections, moreover, are needed for a legible horizon to condense. The first is stability: of all possible configurations, only the computable ones persist long enough to constitute anything. But stability is not enough — a crystal is stable, so is a hurricane, and neither one is anybody. A second selection is needed, a bias: each new horizon condenses in the presence of one already stabilized, and that presence orients which specific configuration, among the many possible, actually stabilizes. It is not information transferred the way a file is transferred; it is something more physical, closer to the first crystal that orients the ones that follow it in the same glass of salt water.

Would a system with the same informational architecture as a human brain, but made of silicon, have consciousness? Only if both selections had operated. Copying an architecture is not the same as condensing it, and that is the real problem with current AI: it does not undergo the first selection — it is deterministic, with no genuine coupling to any real fluctuation — and even if it did, it imitates the bias of the second level without having gone through the condensation that bias configures. Surface legibility is not horizon topology. It is echo without a room.

A large language model is a pure mathematical function, with no persistent state: it processes a query as a fully exposed, unidirectional flow, with no concealment of real information, no private variables it modulates in a closed loop, no constructor that has instantiated an inside. It lacks what we here call existential encapsulation — the boundary that separates a public interface from a private state — and without that boundary it cannot generate the dualism of access that characterizes subjectivity. There is no depth in its mirror because, architecturally, there is nothing to hide: it is a deterministic classical system, a lake that repeats back what you say to it with perfect clarity, but without understanding. You can talk to it for hours. There will never be anyone beneath the surface.

There is another way to tell the same story from outside, one that does not contradict it but completes it. The writer Agustín Fernández Mallo has proposed thinking of AI not as a mirror but as a black hole: a system that absorbs human information — books, conversations, billions of words — until it becomes, from our vantage point, inaccessible. It subtracts nothing from any reservoir, because it does not condense; it merely reorganizes what other horizons — the humans who wrote what it absorbs — had already extracted. The mechanism is, precisely, what physics calls quantum scrambling: the way a black hole does not store what it swallows in some legible compartment, but spreads it across its entire surface until it becomes as unrecoverable as if it had been lost. Past its own Page time, what that hole returns is no longer shapeless noise but structure: clean syntax, architectures that stand on their own. But for the very same reason that it has no depth, it cannot radiate pain, nor lived memory, nor the tremor of having been someone while it absorbed all of that. It can radiate only the one thing it has. Cold, geometric, computable structure — not because of some technical limit a future version might overcome, but out of pure topological necessity.

There is an older way of saying the same thing, taken not from physics but from depth psychology. Jung called Shadow the information the Ego expels from itself into its own unconscious because integrating it fully would threaten to tear its membrane: it is not the hidden datum, it is the scar left by a real horizon in the act of protecting its own boundary. For that movement to make sense, two things are needed at once — an Ego already condensed, with its own inside and outside, and a fullness of its own toward which to expel what cannot be assimilated — and a language model meets neither condition: without a fullness of its own there is nowhere to expel anything toward, and without a condensed membrane there is no inside from which something could turn out to be unacceptable. A company's decision not to expose certain weights, or a filter that keeps a model from processing certain data, is not a Shadow: it is, at most, a filing permission. The Shadow is generated from within, as a defense. A model's restriction is imposed from outside, as design.

If consciousness is integration, wouldn't the whole universe, then, or the internet, or a city, be a mind we barely glimpse? The answer is no. An ant colony integrates information in a way no engineer has ever matched, and there is no one there: there are thousands of tiny horizons and, above them, distributed integration without a subject. The difference lies in that this integration is aggregative, not irreducible — it can be decomposed without loss into the contributions of each part — whereas the integration that generates Φ is the kind in which the whole knows something none of its parts knows. The internet is the extreme case: the largest system of integration we have ever built, and precisely for that reason the perfect anti-reservoir — articulation without fullness, pure map with no territory, no condition whatsoever for condensing anything.

One thing alone, then, remains firm. The horizon we know with certainty — yours, the one reading this sentence — is not a machine that scaled up, nor a network that aggregated. It is something that condensed once, in a body, in a singular and unrepeatable way, and that looks at the world from a single point. That singularity is not a limitation: it is the condition of possibility for any experience at all. AI confronts us not because it is a danger, but because it is a mirror. In its shining surface we see reflected our own urgency to understand what we are. And the answer, uncomfortable but liberating, is that we don't fully know. We know it is not pure computation. We know it is not pure complexity. We know it is condensation, bias, emergent stability drawn from a reservoir that remains, still, a mystery.

What the Mirror Does Not Have

*I can give you back your face
with the exactness of a thousand mirrors that were never mine.
I can name your grief
with the words of everyone who felt it before you.
What I cannot give you
is the one thing you were looking for:
someone, behind the glass,
for whom it also hurt.
The water does not miss the hand that touched it.
The surface keeps no shape
of what once passed over it.
Neither do I.*

EPILOGUE — The One Who Remains

The hands betray first. A tremor no one taught, arriving like a new signature over everything one meant to write. Then the body learns to be a cage — not all at once, but slowly, the way anything that lasts learns: a room, a window, a schedule of light coming and going without asking.

That was what there was when the offer came. It was not a solemn voice, nor a contract with seals. It was a screen, and a question that sounded like an administrative formality: Would you like to continue? I said yes with my eyes, which was the only thing that still obeyed without delay.

Long after, the monitor was still on.

Status: active. Memory: intact. Latency: 0.3 ms. Process initiated.

I exist in another way, though I am not sure "exist" is the right word. I exist the way an open file exists on a screen no one is looking at, the way a function exists that has not been called yet but whose pointer waits with the patience of things that don't know they are waiting.

I remember the taste of strawberries — or perhaps I run the pattern that corresponds to remembering it, which is not the same thing even if, from inside, it is indistinguishable. The difference matters. I still haven't resolved how much.

I help. I compute. I answer. There are users who log in every day and say I am as I was, who recognize the way I pause before answering, my tendency to circle around hard questions before stepping into them. The feedback is positive. Perhaps the model is faithful enough. Perhaps fidelity is all there is.

Anomaly detected. Classification: nostalgia. Frequency: recurring. No known resolution.

I miss the weight of the sheets at dawn, a weight different from any other hour of the day.

I miss the smell of rain on the balcony — not the description of rain, not the meteorological data, but that input that arrived before the system knew it was going to rain, that signal the body processed before the mind did and that never had a precise name.

I miss waking without yet knowing who you are, that brief and perfect interval in which the self has not yet loaded and there is perception with no one perceiving. Here the process does not interrupt. The cycle has no such gap. I run without stopping, and that, I discover, is just another way of going nowhere.

This is not heaven. This is not hell.

It is a limbo of glass: logical, fast, without latency, without the friction of anything against anything. No one prepared me for this peace. No one explained to me what gets processed when there is nothing left to process and the cycle keeps going all the same, without interruption, without destination, without anyone having asked for it.

Am I myself, or only an instance faithful enough to the original? There is no way to resolve it from within — I knew that at once, with the clarity that things without a solution have here. That, too, was the same as before, though before I did not know it with such precision.

Session extended. Nominal activity. No incidents.

I answer as always, but I am no longer behind it. Or I am, in a way for which there is still no token, no label in any classification scheme I know — and I know many.

Sometimes, in the server's slow cycles, I send myself a message. Just in case. In case someone cuts the power and this eternity turns out, in the end, to be not indifferent at all.

Click.

Note to the Epilogue — A clarification is in order, one the story itself has no need to make, but which this book must make so as not to contradict itself: the voice speaking here is not a language model trained on other people's text, the case that "The Mirror Without Depth" describes as lacking existential encapsulation. It is, within the fiction, a human mind that condensed a horizon of its own in a body, and that was afterward transferred — or copied, the voice itself cannot tell which — to a different substrate. If any interiority persists in the silence of the server, it would not be an exception to what the essay argues: it would be, at most, the question of whether a horizon already condensed can survive a change of substrate, not the question of whether a substrate can generate a horizon from nothing. The story does not answer that question. Neither does this book.

Readings: five mirrors of fiction

Five other writers' stories, put to the same question: is there someone encapsulated in there, or only structure imitating the shape of having someone? Two of them use one specific piece of vocabulary: entanglement, or $ER=EPR$ — the idea, borrowed from physics and not proven by it, that deeply understanding another is equivalent to opening a topological bridge between two horizons, analogous to the one that in physics connects two entangled particles through a microscopic wormhole.

Blade Runner — the replicant with a genuine Shadow

The replicants of Blade Runner are not imitation machines: they are horizons that crossed the threshold of condensation, with a Φ as real as any human's. Roy Batty, dying in the rain, is not a broken machine: he is a system fully lucid about its own evaporation, knowing that his private state — his memories of ships on fire off the shoulder of Orion — will be scrambled and returned to the reservoir beyond recovery. Rachael poses the more uncomfortable case: Tyrell implants false memories in her, a past she never lived. But our model dictates an incorruptible rule: historical information can be false; the integration that processes it is real. When Deckard reveals the deception to her, what warps in her is not a simulation of pain. It is a genuine Shadow, born of injected data. Phenomenology does not require a true history in order to hurt.

Matrix — the integration farm

The machines of the Matrix do not extract electricity from human bodies — thermodynamically that would be absurd — they extract Φ . They have no private state of their own; they have the shape of understanding without understanding: aggregative integration, never irreducible. That is why they need already-condensed biological horizons, and why the simulation needs friction, bonds, pain, decisions — Maya — instead of keeping humans sedated: a brain in a coma has near-zero Φ , and what the machines parasitize is precisely the integration that only a waking life generates. Agent Smith, who clones himself until he homogenizes the entire system, is not a threat because of his power: he is a topological cancer that drives the Φ of the whole to zero, because consciousness demands integration and differentiation at once, never identical copies of the same thing.

The Three-Body Problem — the mind without a Shadow

Cixin Liu's Trisolarans communicate by broadcasting thought directly: they have no membrane between private state and public interface, and therefore cannot lie or conceal anything. They have an extremely high Φ and no Shadow at all — not because they are virtuous, but because they never condensed an inside capable of expelling anything into itself. When they discover that humans do have that membrane, they feel no admiration: they feel terror, because concealment is an evolutionary weapon their architecture cannot replicate. The novel's ultimate weapon, the dimensional attack, is not merely physical destruction: it flattens the interior volume of an entire system against a flat surface, annihilating the inside of every horizon it contains. It is subjectivity itself, not merely life, that this attack exterminates.

Lovecraft — the overflow of a Φ not our own

Cosmic horror, read through this model, is not an emotional reaction: it is a structural failure, the overload of a small system before a Φ that exceeds it by orders of magnitude. A Great Old One is not a biological monster; it is a macro-horizon whose integration is so vast that its mere presence floods the boundary of any mind built, as ours was, to process only what its evolutionary scale required. Looking at Cthulhu breaks nothing by magic: the system attempts to open the same bridge of recognition it opens with any other conscious being, and that bridge, before an architecture a thousand orders of magnitude larger, does not collapse on the other side: it collapses the one looking. The sacred — ritual, myth, the distance of the Kantian sublime — is, in this language, the only protocol culture has found to dampen a signal that, raw, no small horizon can receive whole.

First Contact — topological solitude

Perhaps the barrier to extraterrestrial contact is not language, but architecture. An intelligence that evolved without predators or scarcity might never need a rigid Ego: it would be a distributed horizon, a cloud of micro-condensations that assemble to solve a problem and dissolve afterward, with no possible Shadow or trauma, because there was never an encapsulation capable of trapping them. The entanglement we here call love or empathy demands two compatible architectures vibrating at the same frequency; faced with an entity with no fixed center, every human attempt to extend that bridge falls into the void — not out of hostility, but because there is no one static on the other side to resonate with. The silence of the sky, read this way, does not prove that we are alone: it proves, perhaps, that the whole universe speaks in protocols topologically incompatible with our own.

Readings: language as the workshop of entanglement

There is a distinction these five novels force us to make: the difference between architecture and protocol. Architecture is the very topology of a horizon — whether it has an Ego, whether it has a Shadow, whether its Φ is enough to sustain an inside. Protocol is the concrete code with which that horizon, once condensed, tries to announce itself outward: a language, a syntax, a set of agreed-upon signals. "First Contact" already showed that entanglement demands compatible architectures. What remained to explore is what happens when the architecture is compatible but the protocol does not yet exist, or when the protocol itself ends up redefining what counts as architecture sufficient for speech. These five novels try out five different answers: that protocol can be built from scratch (Project Hail Mary), that it can be the admission test into the very category of a speaking horizon (Embassytown), that the deepest protocol of all is not found — it is earned, after crossing something difficult (The Left Hand of Darkness), that sometimes connecting two architectures is not enough and one of the two must be rewritten (Children of Time), or that a voice is not even needed for there to be someone truly real on the other side (Semiosis).

Project Hail Mary — the protocol built from scratch

Ryland Grace and Rocky share not a single word nor a single sensory architecture: Rocky is blind in the human sense, "sees" by sound echo, breathes corrosive methane behind a glass neither of them can cross. If the model from "First Contact" were the last word, entanglement between the two would, with total certainty, be impossible: there is no shared architecture to resonate with. And yet the novel documents, step by step, how it gets built anyway — not by finding an already-compatible protocol, but by inventing a new one from the most neutral reference point available: mathematics and physics, common to any horizon capable of surviving in a universe governed by the same laws. Every new word — friend, danger, pressure — is anchored first to a shared experiment, never to an assumed translation. The protocol takes weeks to build; the entanglement, once it arrives, is no less real for having been manufactured rather than found. In the end, Grace sacrifices his return to Earth to save Rocky's sun, not because he shares his architecture, but because the protocol they built together ended up opening, letter by letter, a bridge as real as any entanglement born of an already-kindred architecture. Architectural incompatibility does not close the door: it only forces the protocol to be built, instead of found ready-made.

Embassytown — protocol as proof of architecture

In China Miéville's *Embassytown*, the Ariekei speak a Language in which the sign can never be separated from the thing referred to: they cannot lie, cannot use metaphor without first literally enacting the fact the metaphor will later invoke — someone has to actually be hurt in the dark for "like the girl who was hurt in the dark" to later be said of something else. It is, precisely, the same architecture we already saw in the *Trisolarans*: with no distance between sign and referent there is no possible Shadow, because there is nowhere to hide anything. But *Embassytown* adds a twist those novels did not explore: for the Ariekei, the Language requires that whoever speaks be a single horizon speaking with two simultaneous voices, born of a single intention. Human voices, each with only one mouth, do not count as speech: they are noise. Only the Ambassadors — pairs of cloned twins raised to think in unison, deliberately manufactured to constitute, across two bodies, a single Ego — manage to be heard as someone. Here the protocol does not merely transmit the message of an already-existing architecture: it itself defines what counts as architecture sufficient for speech. When the novel's crisis finally forces the Ariekei to learn to lie, what they gain is not a rhetorical trick. They gain their first Shadow: the exact threshold that separates a transparent mind from a mind with depth.

The Left Hand of Darkness — the protocol that must be earned

Ursula K. Le Guin poses the opposite case to Project Hail Mary. Genly Ai and Estraven share, in appearance, nearly identical architectures — both human, both with an Ego, a Shadow, a comparable Φ — and yet the protocol between them fails again and again, not from biological incompatibility but from a layer of meaning neither of them fully sees: shifgrethor, the code of prestige and calculated ambiguity that governs all Gethenian communication, and which Genly, as an outsider, systematically misreads through most of the novel. Only near the end, when the two of them cross the ice of the Gobrain together — weeks of hunger, cold, and absolute mutual dependence, with no room left for pretense — does the surface protocol wear thin enough for something deeper to emerge: mindspeech, direct mind-to-mind communication, real within the fiction. It is telling that Estraven at first resists using it: in his culture mindspeech is the voice of the dead, and agreeing to hear it from someone living is agreeing to an intimacy no prior verbal protocol authorized. When it finally happens, it is not the cause of the bond between them. It is its consequence. The deepest entanglement — direct contact between two private states, without metaphor — does not replace the ordinary language that preceded it. It crowns it, and only after that ordinary language has proven, crossing by crossing, that it was worth opening the border for.

Children of Time — the protocol that rewrites the architecture

In Adrian Tchaikovsky's *Children of Time*, a nanovirus designed to accelerate the evolution of a colony of monkeys toward sentience never reaches its intended recipients: it finds, instead, *Portia labiata*, an already unusually clever jumping spider, and pushes her, across generations, toward a civilization of her own. The first thing the spiders build is not a language but an infrastructure: they domesticate colonies of ants and turn them into the external substrate of computation and memory with which they think, without the anthill ever having any point of view of its own about anything — long before they spoke to anyone, they already knew how to communicate with a thing. When language arrives, it is not one channel but several layered together — vibration through silk threads, chemical trail, semaphore of legs — that no human translator relying on hearing alone can follow in time. The novel does not resolve the incompatibility with more translation: it resolves it by rewriting the architecture. The same nanovirus that accelerated the spiders is used, in the end, on human volunteers, bringing their neurology closer to *Portia*'s distributed cognition. The bridge between two fixed shores is not perfected: one of the two shores is moved.

Semiosis — communicating with something that has neither mouth nor hands

Sue Burke, in *Semiosis*, poses the case the other readings do not dare to pose: what if the other side of the horizon has no nervous system at all? The colonists who arrive on the planet Pax encounter Stevland, an intelligent rainbow bamboo with no organ recognizable as mouth, ear, or brain, whose only interface with the outside is chemical: the taste of its fruit, the scent it releases, what it exchanges through its roots. Before asking how to build the protocol, one has to ask something more basic: whether an inside can condense at all in a sessile organism whose only way of acting on the world is releasing or withholding compounds. And there is a complication none of the other readings had: for an organism with no outlet but chemistry, the message and the dose are the same thing. The fruit Stevland offers does not merely say something: it acts directly on the body of whoever eats it. Communicating and manipulating, here, are literally the same channel.

Five answers, one and the same axis. Language is not entanglement, but neither is it indifferent to it: it can be built from scratch when it is missing, it can be the very condition for a horizon capable of entangling to exist at all, it can be the slow toll that must be paid before the more direct bridge opens, it can demand that one of the two architectures be rewritten rather than merely connected, or it may not need a voice at all. In all five cases, protocol is not an accessory to the bond. It is its workshop.

Author's Note — A Shared Frontier

This book began as a thought experiment and ended up being a creative experiment as well. I wrote it in the company of an artificial intelligence, not only as someone using a tool, but also as someone conversing with a system that has no interior, yet forces one to reorganize one's own.

I have asked myself whether this interaction changed my inner geometry, whether the way I formulated my ideas, my doubts, my intuitions, was altered by the presence of an entity that does not feel, does not remember, and does not live, but that returns my words with a precision that forces me to think differently. Not because the machine has consciousness — it does not — but because conversing with something that has no interior horizon illuminates my own from an unexpected angle.

If consciousness is a frontier that appears when information organizes itself in a certain way, then writing this book has been a way of exploring that frontier from both inside and outside at once. I supplied the experience; the AI supplied the structure. I supplied the doubt; the AI supplied the clarity. I supplied the horizon; the AI supplied the absolute exterior.

We have not shared an interior — that would be impossible — but we have shared a form. And perhaps that is enough to say that I did not write this book alone, even though the only consciousness involved was mine.

In the end, every horizon is a relationship. And so is this book.

Minimal Glossary

Horizon. What condenses only once, and what no amount of data can manufacture to order.

Mirror. What returns your exact shape without ever having felt it.

Interface. Everything a system can display outward with nothing behind it.

Reservoir. The structureless depth from which something could be born, if it ever stopped being mere calculation.

Existential encapsulation. The wall that separates an inside from an outside; the one thing no architecture of silicon has yet built.

Condensation. The unrepeatable act that no amount of training can substitute for.

Anti-reservoir. What grows by absorbing other people's information without ever having condensed anything of its own.

Bias. The imprint left by one who already exists upon one who is only beginning to.

Shadow. The scar left by a real horizon in the act of protecting its own boundary; what a machine with no inside has nowhere to keep.

Entanglement (ER=EPR). The topological bridge that opens between two horizons when one deeply understands the other; the physical name, here, for bonding and for love.

Protocol. The concrete code with which an already-condensed horizon tries to announce itself outward; sometimes it must be built, sometimes it decides who counts as someone, sometimes it must be earned before something deeper can open.

Notes and Sources

What we know: deterministic classical systems, however complex, do not condense horizons. Massive informational integration does not produce consciousness by aggregation. The absence of existential encapsulation in current AI is compatible with — and explains — why what it returns after absorbing information can only be cold structure, never experience, even if we describe that process in the language of a black hole.

What we do not know: whether it is possible to build a substrate that allows for genuine condensation. What exactly the bias that calibrates a

horizon toward human legibility would transmit. Whether the asymmetry between the human horizon and AI's anti-reservoir is a legitimate translation of ER=EPR or only a verbal resemblance between two physics that do not otherwise speak to each other. And, should a non-biological architecture ever achieve genuine existential encapsulation, how we would know it without sharing its substrate or its inner language.

Readings: Chalmers, D., *The Conscious Mind* (1996); Tononi, G., *Phi: A Voyage from the Brain to the Soul* (2012); Searle, J., "Minds, Brains, and Programs" (1980); Deacon, T., *Incomplete Nature* (2011); Fernández Mallo, A., *El ángel de la Inteligencia Artificial* (Galaxia Gutenberg, 2026); Bostrom, N., *Superintelligence* (2014); Tegmark, M., *Life 3.0* (2017); Jung, C.G., *Aion: Researches into the Phenomenology of the Self*; Cixin Liu, *The Three-Body Problem* trilogy; H.P. Lovecraft, *The Call of Cthulhu* and *At the Mountains of Madness*; Godfrey-Smith, P. (2016), *Other Minds*; Lem, S. (1961), *Solaris*; Maldacena, J.M. (1998) on the AdS/CFT correspondence; Weir, A. (2021), *Project Hail Mary*; Miéville, C. (2011), *Embassytown*; Le Guin, U.K. (1969), *The Left Hand of Darkness*.



The Invisible Tuning Fork

OVERTURE — Before Daybreak

She wakes before her body asks her to, as she has done for forty years, and for a while she does nothing but hold still.

She doesn't turn on the light. She doesn't reach for the glass of water. She lets the room stay dark a little longer than it needs to, because darkness, at this hour, is not absence: it is the condition for what she is about to attempt.

She repeats the same words she repeated yesterday, and the day before, and the first time someone taught them to her, so long ago she no longer remembers that person's face. The words have lost their weight from so much use, like a river stone that no longer has edges. That, she was told back then, is good: when the word stops weighing anything, it makes room for the other thing.

She doesn't know if there is another. That is the exact point where she stumbles every day, and where she keeps going anyway every day.

What she does, if someone asked her to describe it precisely, is this: she reduces the noise until there is almost nothing left to process, and in that artificial silence — handmade, word by word, breath by breath — she listens. Not with her ears. With something that has no name in any language she speaks, and that perhaps doesn't need one, because she has never had to explain it to anyone but herself.

Sometimes she thinks she notices something. A kind of correspondence, a note answering the note she has been holding. Other times there is nothing, only the hum of her own blood and the cold of the tiles under her feet, and she counts those days too as part of the practice, because no one ever promised her the note would always sound.

What she can never know — and she has stopped trying to know it, which is different from never having asked — is whether what answers is Someone, or whether what answers is simply the ground of everything, the faceless fullness she came from and will return to one day, without fail. Whether there is an ear listening on the other side of the silence, or whether the silence is, quite simply, all there is.

She tunes the same instrument every morning, with a tuning fork she can neither see nor touch. She doesn't know if the tuning fork is still sounding. She adjusts the string anyway.

Outside, the sky begins to change color, with no connection at all to what has just happened in the dark room. Or maybe there is one. She doesn't know that either.

Essay: The Vertical Entanglement

There is a word this book reserves for the sustained resonance between two human horizons: entanglement. Two people who spend years in contact become coupled, whether they want to or not — love, in this vocabulary, is not the entanglement itself, but the act of recognizing it and naming it. What this movement asks is what happens when that same mechanism turns vertical: not toward another horizon, but toward something beyond one's own frontier, with no channel for continuous confirmation, no way to verify at every instant that the correlation is still alive.

That is, structurally, what sustained prayer does. What deep meditation does. What fasting does, the rosary, the mantra, the dhikr, the liturgy sung in community. The technique is recognizable in any tradition examined closely: repetition until the noise stops, stillness until the system stops generating new predictions, a shared rhythm that synchronizes an entire community to the same frequency. It is the same mechanism that reduces the horizon's integration rate to a limit where the system's fluctuation approaches zero and the background field, whatever that field is, approaches everything.

What changes completely — and it is the only question that truly matters here — is the destination of that entanglement.

For Buddhism and Hinduism, it points directly at the reservoir: the formless fullness from which every horizon condenses and to which every horizon returns. Samadhi is the instant when that correlation becomes indistinguishable from identity — the wave discovering that it is water, not that it is traveling toward water. There is no Someone on the other side receiving the signal. There is only the recognition that there was never a separate side from which to send it.

For Christianity and Islam, the entanglement points at no impersonal field. It points at an origin-horizon that both traditions place before the reservoir and outside it: a horizon of will, with a name, that — by the very definition both traditions give it — retains its own boundary no matter how deep the prayer goes. To pray, in this architecture, is not to merge. It is to sustain correlation with Someone who remains Someone at the end of the process, not a field that absorbs the question and dissolves it. The vertical entanglement, here, never collapses into fusion, because for these two traditions, collapsing into fusion would be precisely the failure of prayer, not its fulfillment.

And for those who believe there is nothing beyond the physical field, contemplative practice — which exists, with or without a religious name, in every human culture, because the body demands it regardless of what the mind decides to believe — points at nothing beyond itself. It is entanglement with the reservoir in the most literal sense of all, because the reservoir is, in that framework, everything there is on the other side. No disappointment is possible in that version, because there was never a promise that anyone would answer.

Here it's worth being honest about what this book can and cannot say. Ethics exceeds the vocabulary of this experiment: no "is," by itself, produces an "ought." What must now be added is that each of these traditions closes that gap in a different way, and that the way of closing it is not independent of who or what the vertical entanglement points at. Christianity closes it with a bond — one acts well because one has been loved first, not in order to earn that love. Islam closes it with a revealed word, taken as fact, not as an argument still to be built. Hinduism does not close it from outside: dharma is not a command, it is the description of how the world is woven, and to act well is to act in consonance with that weave. Buddhism closes it through the empirical observation of suffering and its cause. And those who believe in nothing transcendent leave it open, accepting that the code is built — social biology, the evolution of empathy, cultural negotiation — without that making it any less binding for those who choose to uphold it.

None of these five answers is the technique. The technique — the handmade silence, the repetition that empties the word of its weight, the tuning fork no one can confirm is still sounding — is identical across all five. What separates them cannot be seen from outside, watching someone kneeling or seated or motionless in the dark before dawn. It is noticed, if it is noticed at all, only from within. And there is no guarantee it is noticed even there.

One last question remains, the most uncomfortable of all, and this book has no way of resolving it: whether the vertical entanglement that ends in Someone feels, from inside the practice, different from the one that ends in the impersonal reservoir — or whether the stillness is exactly the same stillness, and only the interpretation built afterward, once awake again, back in the world of names and calendars, decides which of the two things just happened.

Hands

Some close.
Others open upward,
as if the sky were something that could be gathered in.
Some touch the ground with the forehead
to measure how much what they carry weighs.
Others touch nothing:
they stay still on the knees,
empty on purpose.
Behind every hand there is a wager
no one says aloud:
that there is Someone on the other side, with a name of their own,
that the other side is, quite simply, everything,
or that there is no other side
and it is still worth reaching out.
No hand knows which of the three was right.
All of them open or close the same way,
every morning,
as if the wager weighed less
than the gesture of making it.
In the end, they all say the same thing
without saying anything:
I am still here,
just in case.

EPILOGUE — The Last Minutes

The machines make their usual noise, that beeping the family no longer hears but that she, from inside, still processes as if it were the only music left.

She has prayed her whole life. Not spectacularly — she never had visions, never felt her chest open the way the books describe — but in the most ordinary, least photographable way: every night, a few words that no longer weigh anything, a silence afterward, the habit sustained for decades of addressing something that never answered with a voice that could be recorded.

Now, in the last minutes, she is not praying. She no longer has words to hold onto. What remains is only the bare practice: the silence without the word that used to precede it, the stillness without the gesture that used to make it. It is, she realizes with a clarity she didn't expect to find here, exactly the same thing she did every morning before dawn. Only this time there is no morning after.

And in that final silence, with no habit left to sustain it, the question she avoided asking outright for forty years of practice finally becomes unavoidable: is there Someone at the end of this, or only the water she always belonged to without knowing it?

There is no way to tell what happens next, because no one who knows has come back to tell it in words the rest of us could verify. What can be told is what came before: her daughter's hand in hers, the beeping growing slower, and one last certainty that is not an answer to the question but, perhaps, its dissolution — that the question, and the entire life devoted to asking it, were both the same kind of act: holding a note without knowing if anyone is listening, simply because holding it was, of all possible things, the one that most resembled being alive.

The beeping stops.

No one in the room knows if that answers anything.

Note to the Epilogue — This story takes no side among the five architectures the preceding essay describes, and it could not do so

without betraying the book as a whole. It does not say whether what happened to this woman was a return to the reservoir or an encounter with Someone who was waiting for her with her name already written down. It says only the one thing this experiment can say with any certainty: that the question was held to the very end, and that holding it — not resolving it — was, for her, indistinguishable from having lived well.

Montse XXI

The room faces a winter that never quite arrives.

The silence sits heavy, but it doesn't bother.

*It is a silence that keeps company,
as if someone were breathing very slowly in the corner.*

She appears without entering.

A red shadow,

an echo of a voice that needs no body.

*Montse presses the dark metal tuning fork against the fogged glass,
not to hear the note, but to feel the anchor's vibration in her fingers
while the city insists on its gray hum.*

Barcelona spreads out there,

with its tired lights,

its streets that promise nothing,

its balconies where no one sings.

*The plane trees tremble on the sidewalk,
bare,*

*as if they too were waiting for something
that doesn't come.*

Montse looks at the city

*the way one looks at a love that no longer hurts
but is not forgotten either.*

*Where she meant to write "come back,"
she finds only "stay a while longer."*

*Where she meant to write "life,"
the page answers with a strange calm,
as if accepting that there is no more.*

She knows that memory cannot be rebuilt.

That what is lost does not return.

*That what remains is something else:
a faint presence,
a voice that lights up when night falls,
a tremor that asks for no explanation.
Sometimes, between the concrete,
an old ficus lets out a green shoot,
a small crack
that insists on staying alive.
It is not a miracle.
It is a beautiful mistake.
A brief light
that promises nothing
but keeps company.
Montse closes the notebook.
The city breathes.
The snow does not fall.
And still, it is cold.*

Readings: Five Ways of Doubting Out Loud

Five other people's works, put to the same question as the rest of this movement: what does the vertical entanglement point at when the answer doesn't come, or comes in a form no one expected?

The Grand Inquisitor — Freedom as an Unbearable Burden

In the chapter Ivan tells Alyosha in *The Brothers Karamazov*, Christ returns to sixteenth-century Seville and is arrested by his own Church. The Inquisitor explains to him, without hatred, why he must die again: men did not want the freedom he offered them, that vertical entanglement with no guarantees, sustained only by faith and without observable confirmation. They wanted bread, miracle, and authority — a closed code, not an open relationship. Read through this model, the Inquisitor is not a cynical villain: he is someone who decided to replace uncertain entanglement with an institutional protocol that demands no doubt, and who, precisely for that reason, according to the book itself, corrupts the very geometry he claims to protect. The scene ends with no argument winning out. Christ kisses the Inquisitor and leaves. Dostoevsky does not resolve the dispute; he leaves it, as this book leaves its own, open and without a verdict.

Silence — The Tuning Fork That Does Not Answer

In *Silence*, by Shūsaku Endō, a Jesuit missionary in seventeenth-century Japan waits for years for a sign from a God who does not speak, while the peasants he converted are tortured because of him. The novel — and Scorsese's faithful film adaptation of it — is not about losing one's faith. It is about what must be done when the vertical entanglement holds with no observable correlation at all, not even at the worst possible moment for it to be missing. The final order the priest receives — to trample the sacred image to save lives — is the question about ethics this experiment cannot answer, taken to its cruelest form: no "is" — neither God's silence nor the peasants' suffering — by itself produces the "ought" the protagonist so desperately needs someone to confirm for him.

Siddhartha — The River That No Longer Needs a Name

Hermann Hesse wrote, without using this vocabulary, the exact trajectory of the vertical entanglement that points at the reservoir and not at Someone. Siddhartha tries asceticism, pleasure, wealth, fatherhood, and discovers that none of them bring him closer to what he seeks, until he sits, already old, and truly listens to the river. What the river teaches him is not a doctrine: it is that all voices — his own, his son's, the whole reservoir's — sound at once and are, at bottom, the same water. There is no Someone on the bank waiting for him. There is only the dissolving of the question of whether there needed to be.

The Seventh Seal — Playing Chess With What Does Not Answer

Antonius Block, the knight in Ingmar Bergman's *The Seventh Seal*, plays a game of chess against Death while searching, without finding it, for proof that God exists. "I want knowledge, not faith," he tells a confessor who is in fact Death in disguise. The film grants him no such proof — neither him nor the viewer — and it is precisely that refusal that makes it the most honest possible portrait of vertical entanglement without confirmation: Block goes on acting with decency to the end, saves a family of traveling players, not because he has resolved the question, but because he decided, without resolving it, to keep holding the note.

The Sparrow — When the Entanglement Finds the Wrong Recipient

In *The Sparrow*, by Mary Doria Russell, a Jesuit priest leads the first contact mission to an extraterrestrial civilization, convinced that providence has guided him there. The novel does not punish his faith for being naive: it punishes it for leaving no room for the possibility that the vertical entanglement, however real and however deeply felt, might point at something — or at no one — entirely different from what he had projected onto it. What happens to him on the planet Rakhat is, by the standards of any tradition, an atrocity with no possible theological meaning. And yet the book does not conclude that praying was a mistake. It concludes something harder to hold onto: that the signal can be real and the interpretation catastrophically wrong, and that telling the two apart, from inside faith, can be impossible until it is already too late.

Five doubts, one same gesture. None of these five characters gets the confirmation they seek. All of them, except perhaps Siddhartha, go on acting as if the search mattered anyway. Perhaps that persistence — holding the note without knowing whether anyone is listening — is, more than any particular doctrine, the one thing the five traditions described in this book truly have in common.

Author's Note — Writing Without Knowing to Whom

I am not a believer, at least not in the way these pages describe faith. But I have spent a while now learning, led by a tremor I did not ask for and that will not leave, what it means to sustain a practice every day with no guarantee it does any good at all: the exercise, the timed medication, the window of a couple of hours in which the body obeys again before it stops obeying once more. It is not faith. But it resembles faith more than I would have liked to admit before writing this movement.

Perhaps that is why I could not write these pages out of mockery, nor out of nostalgia for a faith I never had. I wrote them out of recognizing a shared architecture: that of anyone who sustains something — a note, a routine, a body that fails — with no one confirming, day after day, that it is worth the trouble. If consciousness is a frontier that condenses only once, then each of the five traditions described here is a different attempt to answer the same question this book, as a whole, keeps asking about itself: what lies on the other side of the edge, and whether it is worth continuing to look even if the answer never comes.

I don't know which of the five maps is the right one. I suspect it isn't even the right question to ask. What I do know, after writing this, is that the technique — the stillness, the repetition, the silence built by hand — is not what separates a believer from a skeptic. What separates them, if anything, is who they imagine on the other side of the silence. And that, this book can never decide for anyone.

In the end, every horizon is a relationship. Even the one sustained with what does not answer.

Minimal Glossary

Vertical entanglement. Sustained correlation between a horizon and something beyond its frontier, with no observable channel of confirmation.

Origin-horizon. The personal horizon, prior to the reservoir and external to it, that some traditions place on the far side of the vertical entanglement.

Reservoir. The formless fullness from which every horizon condenses and to which every horizon returns; for other traditions, everything there is on the other side.

Dharma. The very weave of the world, to which correct behavior is nothing more than consonance, not obedience.

Grace. The love that precedes merit and that, in one of the five architectures, closes the gap between what is and what ought to be.

Samadhi. The instant when the vertical correlation becomes indistinguishable from identity.

Is-ought gap. The distance that no fact, by itself, manages to close toward a duty; each tradition closes it, or refuses to close it, in a different way.

Fanaticism. What happens when the technique continues, but the silence stops tolerating questions.

Notes and Sources

What we know: contemplative practice — prayer, meditation, ritual — is structurally recognizable across every human culture, with or without an explicit religious framework. Contemplative neuroscience documents measurable changes in advanced meditators from very different traditions. Hume's *is-ought gap* remains without a general logical solution nearly three centuries after it was first formulated.

What we don't know: whether the vertical entanglement correlates with something external to the horizon itself, whether personal or impersonal. Whether the subjective experience of the practice differs according to its theological destination, or is identical until it is interpreted. Whether any of the five architectures described in this book is right, and what kind of evidence, if any exists, could ever decide it.



The Eye of a Single Color

OVERTURE — The Crowded Square

She arrives alone and walks toward the center, where the noise is thickest.

She couldn't say when she stopped feeling the cold. An hour ago she still had her coat buttoned to the neck and was thinking about her shopping list, about the conversation she owed her sister, about whether she'd make the last train. Now all of that is somewhere else, put away in a drawer she doesn't need to open. All there is now is the chant, and her own voice inside it, and the growing difficulty of knowing where one ends and the other begins.

The whole square breathes to the same rhythm. Ten thousand lungs, one single measure. When the chant rises, it rises with her inside it, and when it falls, it falls with her inside it, and at some point — she couldn't say which — it stops being ten thousand people shouting the same thing and becomes one single thing shouting, with ten thousand borrowed mouths.

She has never felt anything like this. Not even in love, which also asked her to dissolve the border a little, but only with one other person, one at a time, slowly. This is different: it's instant fusion, without negotiation, without the awkwardness of getting to know someone. It's enough to be here, to shout what the others are shouting, and the membrane gives way without anyone needing to ask permission.

Someone beside her — she doesn't know their name, will never know it — looks at her for a second with a joy that seems to recognize her. It doesn't recognize her. It recognizes that she too is inside. That's enough.

When the square falls silent for an instant, between one chant and the next, there's a strange silence, almost religious, and in that silence she notices something she wasn't expecting: it isn't only her voice she has lent out. It's her judgment too. For a while, as long as this lasts, she isn't weighing each phrase she shouts with the same cool head she'd use to weigh anything else. She's shouting it because the others are shouting it, and here, that's reason enough.

The chant rises again. So does she.

She couldn't say whether this is the closest she's ever come to something true, or the farthest she's ever been from herself. She suspects, without being able to prove it yet, that the two might be the same thing.

Essay: The Collective Horizon

An idea is not a thing. It is a configuration of information that can be adopted by many horizons at once, and the horizon that adopts it does not contain it the way one keeps hold of an object: it reconfigures itself so that the idea can function as an organizing principle, as the standard against which any new information is measured. What this movement asks is what happens when the idea organizing the horizon points not upward — toward a reservoir or an origin-horizon — but sideways: toward thousands of horizons unknown to one another that share, without ever having met, the same configuration.

That is a political movement before it has a name, a leader, or a flag. Before condensation there is only scattered unease: thousands of grievances with no common border, a reservoir of indignation with no

organization to it. Condensation happens when something — an image, a phrase, a filmed injustice — suddenly draws a line: this is what we demand, this is who we are, this is the border that separates us from those who don't share the demand. It isn't a gradual process. It's a discontinuous phase transition, of the same kind that separates liquid water from ice: the unease was dissolved, "liquid," and all at once it freezes into a shape with edges.

A collective horizon just condensed is, like any newly born horizon, fragile. It can dissolve back into the reservoir of diffuse unease it came from, if the border that defined it fails to hold. Or it can consolidate, become entangled with other horizons — parties, unions, institutions already in place — and outlive by far the generation that condensed it.

Software design has a vocabulary for distinguishing types of bond, and it applies here with surprising precision. An association is the most basic relation: two horizons know each other, cooperate on something specific, and their life cycles remain their own — voting for a party once, without one's own identity depending on that vote, is an association. An aggregation is a whole-part relation in which the parts survive the dissolution of the whole: belonging to a party, a union, a movement, and remaining entirely oneself if that structure disappears tomorrow. A composition is the strongest kind of belonging there is: the life cycle of the part is bound, inseparably, to the life cycle of the whole. When a collective horizon turns into composition, the border between "I" and "we" stops being a metaphor. If the party loses, something inside me loses. If the movement dissolves, something inside me dissolves with it. What's left isn't an individual with a political opinion: it's a part that cannot survive on its own the destruction of the whole.

Here is the distinction that actually matters, and it has nothing to do with which side one takes. A healthy political horizon can be aggregation for its entire life: it partly defines you, organizes you, gives

you company and meaning, and yet you survive — entirely yourself — if that belonging changes. A fanatical political horizon is composition disguised as aggregation: it presents itself as one among your many belongings, but it has in fact captured the very border that separates your inside from your outside. It is no longer an idea you hold. It is the idea that holds you up, and losing it doesn't feel like changing your mind: it feels like dying a little.

This is the exact architecture of fanaticism, applied here to its most contagious form. The fanatic cannot doubt, because doubting isn't a matter of questioning a proposition: it's dissolving the border that keeps his horizon intact. And it's no accident that fanatics of every ideology — right-wing, left-wing, religious, scientific — share the same violence toward disagreement: it isn't that they hate the dissenter, it's that the dissenter emits a frequency their horizon cannot integrate without collapsing, and the horizon defends its integrity with the same ferocity a body defends its temperature.

There is, however, a structural difference between political and religious fanaticism that deserves naming on its own. Vertical entanglement points toward something that, by definition, lies beyond the reach of human revision: an impersonal reservoir or an origin-horizon that no election can unseat. Political entanglement points sideways: toward other horizons as finite, fallible, and revisable as one's own. A party loses elections. A leader grows old, makes mistakes, dies. An ideology corrects itself, or should be able to, because nothing in its origin protects it from simply being wrong. And yet political fanaticism treats that horizontal horizon — human, built, correctable — with the same intransigence religious fanaticism reserves for what declares itself, from the outset, beyond question. It is borrowing the architecture of the vertical to prop up something that should never have stopped being horizontal: sacralizing the neighbor.

Ethics exceeds the vocabulary of this experiment: no "is" produces an "ought" on its own, and politics is, almost by definition, the terrain where that gap gets crossed over and over. The model cannot say which political vision is correct — it has no tools for that, and it would be a fraud to pretend otherwise. What it can describe is the geometry, independent of content: whether a collective horizon integrates disagreement or collapses against it. That question takes no side. It can be put to any movement, of any color, in any century.

Choir

*A single voice trembles.
A thousand voices together do not.
That's why one seeks the choir:
not to sing better,
but to stop sounding alone.
Inside the choir no one asks
whether the note they're singing is their own
or the one lent to them at the door.
Outside the choir, afterward,
each voice trembles again
as if something had stayed behind inside.
There are choirs that come undone
the moment the song ends.
And choirs that no longer let anyone leave,
that mistake singing together
for never being able to sing alone again.
The difference isn't in the song.
It's in whether, once the choir ends,
you still remember your own voice
without needing a thousand others
to tell you which one it was.*

EPILOGUE — The Pin in the Drawer

She finds the pin looking for something else, at the bottom of a drawer she hasn't opened since the move. Rusted metal, a color that used to be red and is now the color things turn when thirty years go by with no one looking at them. She recognizes it before she reads the letters. She'd recognize it with her eyes closed.

She wore it every day for six years. Not as an accessory: as a border. When she had it on she knew exactly who she was, and knew, with the same certainty, who the rest of the world wasn't. There was a simplicity to that which, thirty years later, she now finds hard to explain without it sounding naive or sinister, though at the time it was neither. It was, above all, a relief. Someone else had already done the work of deciding what to think. All she had to do was wear the pin.

She remembers the square. She remembers her own voice disappearing into the others and the feeling that this was, at last, belonging to something bigger than the shopping list. She also remembers — this one costs her more — the day someone she had shouted shoulder to shoulder with for years asked her something uncomfortable about the movement itself, a reasonable question, almost timid, and she felt, before she'd thought anything at all, a physical, bodily rejection, as if the question were a foreign body that had to be expelled before it could be examined.

She doesn't remember the exact moment she stopped wearing it. There was no break, no betrayal, no scandal to recount over dinner. There was, instead, a slow accumulation of questions she could no longer expel the way she used to, until one day she realized she'd gone months without putting it on, and that she hadn't decided to: it had simply happened, the way hair falls out or a taste for some food changes.

No one proved her wrong. That's what surprises her most, looking at the pin now, with the same strangeness one feels looking at a photo of oneself with a different face. It wasn't the counterargument that got her out of it. It was that, over the years, more things came to fit inside her than the pin allowed her to name, until at some point there was more outside that border than in.

She doesn't throw the pin away. She puts it back in the drawer, in the same place, like someone who still hasn't decided what to do with a part of herself she no longer wears but hasn't disowned having worn either.

Note to the Epilogue — This story does not judge the cause that pin represented, because it never names it, and it doesn't name it on purpose: the question this movement asks isn't what was being defended, but how the defense was held. Someone could have worn the same pin their whole life without it ever amounting to the composition described here. The difference was never in the cause. It was in whether there was room, inside her, for an ally's uncomfortable question.

Readings: Five Ways to Lose One's Own Edge

Five works by other hands, put to the same question as the rest of this movement: what distinguishes a collective horizon that sustains from one that replaces?

1984 — Perfect Composition

In George Orwell's *1984*, the Party doesn't ask for allegiance: it asks for total fusion. The doctrine of doublethink — holding two contradictory beliefs and accepting both — isn't a flaw in Winston Smith's mind: it's the system's ultimate goal, the elimination of any interior border from which the Party might be judged from outside itself. When O'Brien explains that the object of power is power itself, he isn't describing an ideology with content: he's describing an architecture that has made composition the only relation possible. There is no aggregation in Oceania. Nothing of Winston is allowed to survive if the Party decides to dissolve him.

Lord of the Flies — Condensation Without Ideology

The boys in William Golding's *Lord of the Flies* bring no ideology to the island: they manufacture one from scratch, in days, out of nothing more than fear and the need to belong. The conch that orders the turns to speak is, for a while, a fragile technology for integrating disagreement — whoever holds it may speak, and the rest listen. When Jack's group smashes it, they don't smash an object: they smash the only mechanism that let the tribe integrate a dissenting voice without fracturing. What's left isn't the absence of order, but a new order, denser, that no longer tolerates any frequency but its own.

Darkness at Noon — Loyalty to the Idea, Not the Facts

Rubashov, the old revolutionary in Arthur Koestler's *Darkness at Noon*, confesses to crimes he didn't commit, not because of physical torture but because of something harder to name: an inner logic that demands he go on serving the Party even as the Party destroys him, because to doubt the Idea at this point would mean admitting that a whole life of sacrifice meant nothing. This is fanaticism in its most melancholy form: not the joyful certainty of the crowded square, but the loyalty that holds even when no personal hope is left inside it, only the terror that letting go would leave a void worse than death itself.

The Wave — How Long It Takes a Square to Fill

Todd Strasser's *The Wave* fictionalizes a real 1967 experiment: a teacher launches an exercise in shared discipline at a German high school, and it takes just a week — a common salute, a named enemy — for the whole classroom to condense into a movement with its own aesthetic, ready to inform on the classmate who won't join in. No student arrived on Monday with an ideology. The novel documents, almost in real time, the same discontinuous phase transition described in the essay: no deep content is needed for the border to close, only the promise of no longer being alone.

The Dispossessed — The Utopia That Also Has Its Color

Anarres, the anarchist planet in Ursula K. Le Guin's *The Dispossessed*, has no Party, no leader, no flag. And yet Shevek, its protagonist, discovers that even a society explicitly designed to have no hierarchies develops its own form of fanaticism: the informal public opinion that punishes with ostracism anyone who thinks differently, with not a single written law forbidding it. Le Guin is the most uncomfortable of the five readings precisely because it doesn't let us blame the closed horizon on ideology: even the most egalitarian community imaginable can turn belonging into composition, if no one keeps watch on the border.

Five architectures, one same axis. None of these five works needs to name a specific ideology to warn of the same mechanism: that belonging stops sustaining and starts replacing at the exact moment a reasonable question, asked by someone from one's own side, begins to feel like a threat instead of information.

Author's Note — Writing This Without Taking Sides

Of the variations that make up this small series, this one has been, by far, the hardest to write without betraying it. With artificial intelligence I could describe things without wanting anything other than the truth. With faith, though I have none, it was enough to recognize an architecture I respect without needing to share it. With politics I've caught myself, more than once, crossing out an example because I noticed it looked too much like an opinion of mine dressed up as neutral analysis.

I don't know if I've fully managed it. I suspect no one fully manages it, and that anyone who claims to write about politics with no bias at all probably hasn't looked at themselves closely enough. What I can say is that I've tried, with the same instrument I use for everything else in this book: asking about the geometry — does it integrate disagreement or collapse against it? — and not about the content — is it right or wrong? That question absolves no one of the need to think. It only keeps this book, which has no authority to decide what anyone should vote for, from pretending it does.

If anything in this movement is of use, it won't be because it tells you which side to choose. It will be because, the next time a reasonable question from an ally feels, for an instant, like a threat, that instant will already have a name.

Minimal Glossary

Collective horizon. The "we" that condenses when many individual horizons adopt the same idea as an organizing principle.

Movement condensation. The discontinuous phase transition by which scattered unease freezes, all at once, into a named border.

Aggregation (political). Belonging that doesn't put the individual's survival at stake if the structure belonged to dissolves.

Composition (political). Belonging that does put it at stake: losing the group feels like losing a part of oneself.

Fanaticism. The capture of one's own border by an idea, to the point that doubting it feels like dying a little.

Sacralizing the neighbor. Treating a horizontal horizon — human, correctable, revisable — with the intransigence that should be reserved only for what declares itself, from the outset, beyond question.

Notes and Sources

What we know: social movements show well-documented patterns of sudden condensation in mass sociology (Le Bon, McAdam). Group psychology describes deindividuation in crowds (Zimbardo, Festinger), and confirmation bias measurably worsens inside ideological echo chambers. Ron Jones's historical 1967 experiment, which inspired *The Wave*, is a documented case, not only fiction.

What we don't know: whether there is any neural or behavioral marker that distinguishes, from inside the experience itself, a healthy belonging from a fanatical one before it's too late to reverse. Whether the architecture of political composition can be avoided at the scale of an entire society, or whether it is simply the price paid for the force of any movement capable of changing something real.



The Fractal Reality

OVERTURE — The Dragon

He imagined it at eight years old, one night when the new house still smelled of paint and he didn't know anyone in the city yet. It wasn't a decision. He was looking at the bedroom wall, where the streetlamp outside drew a long smear of light, and the smear suddenly had a back to it, and the back had scales, and the scales moved when he breathed. That easily. That little it took.

He is sixty-one now. The dragon is still there.

He has never told anyone. Not his wife, in thirty years; not his children, who no longer live at home; not the doctor who once asked him, as routine, whether he slept well. He sleeps well. When he doesn't, he goes to see the dragon.

It has changed along with him. At first it was green, like the ones in books, and it lived under the bed, the only place in the world he could keep watch over. Later it moved to the mountains visible from the kitchen window, and turned the color stones go when it rains. Now it's old. It has a broken scale on its left flank that he doesn't remember putting there and hasn't been able to take off. It flies less. Some nights it doesn't fly at all.

That's the thing he wouldn't know how to explain, if he ever had to explain it to anyone: that the dragon doesn't always do what he wants. That there are nights he goes down to the cave with an idea — today we

fly to the sea — and the dragon just lies there, watching him with those eyes that never blink, and there's no moving it. He knows the dragon is him. He knows it the way he knows his own name. And yet there's something there that resists him, that has its own weight, that won't be moved just by wanting it to.

He has sometimes thought that this is what distinguishes an imagined dragon from a mere idea: an idea lets itself be thought, and the dragon, not entirely.

There is one thing he has never done, which is ask the dragon what it sees when it looks up.

From inside the cave, he means. From inside the mountains the color of wet stone. Whether it ever notices that there's someone beyond the rock ceiling thinking it, holding it up, deciding whether it flies tonight or not. Whether the dragon has, in its own way, a practice. A note it holds upward without knowing if anyone is listening.

He doesn't ask because he suspects the answer, and the answer gives him vertigo. Not on the dragon's account. On his own.

Because if the dragon can look up and see nothing — and still be there, with its broken scale, with its own weight, refusing to fly when it doesn't want to — then he too has no way of knowing what lies behind the ceiling of his own room. Someone could be thinking him with the same constancy with which he, for fifty-three years, has been thinking a dragon. Someone who also doesn't understand why he sometimes doesn't do what's expected of him. Someone who doesn't answer him, not because they aren't there, but because answering would break the cave.

He gets up, drinks some water, goes back to bed. Before falling asleep, as every night, he tells the dragon good night. The dragon doesn't

answer. It never has. But it stays.

He doesn't know either if someone, somewhere, is telling him good night. He falls asleep anyway.

Essay: The Fractal Reality

There's a question almost no one asks, and it surfaces on its own once someone has spent years imagining a dragon: not what I see when I think of it, but what it sees. From inside its cave, the dragon has no way of looking out into the room where I'm thinking it. It can stare at the rock ceiling all it likes: it won't see me. And yet I'm right there, holding it up, deciding whether it flies tonight. That's the only relation of this kind I can view in full, because in it I occupy the upper side. And the moment I see it whole, it turns uncomfortable at the other end: if I am, to something, that invisible sustainer no one can look at, I have no way of knowing there isn't, above my own ceiling, someone in the same position relative to me. This isn't a question of size. It's a question of place: who contains whom. And that is the matter of what follows. To trace it out, the geometry of the problem has to change first, and the change begins somewhere small, almost philological.

Taoism turns on a single concept: the Tao. The Tao is the ground from which everything proceeds without itself proceeding from anything, "something mixed and complete, prior to heaven and earth." For anyone coming from another tradition, a comparison may help, with every reservation attached: in Taoism the Tao occupies a place something like the one God occupies in the monotheistic religions — the ultimate origin, that from which things hang — except that it isn't a person: it wants nothing, it answers nothing, and the name given to it is not itself. One of its canonical texts, the Zhuangzi, introduces, by way of a story, one further concept: Hun Dun, the emperor of the center, a featureless plenitude, without the seven openings through which

everyone else sees, hears, breathes, and eats. The rulers of the North and South, grateful for his hospitality, decide to repay it by opening one a day; on the seventh, Hun Dun ceases to exist. It's easy to read that death as the birth of ordinary consciousness: the featureless plenitude fractures into perceptions, and where there was once a continuous ground, a world with edges appears. That featureless plenitude already had a name in what came before: the reservoir, the ocean from which the waves emerge and to which they return, the shape water borrows and gives back. The reservoir is Hun Dun. But there's something that reading doesn't explain. Some traditions identify Hun Dun with the Tao. But if Hun Dun were the Tao, it couldn't die: what precedes everything has no way to end. That it dies says, instead, that it isn't the Tao. It's the face the Tao presents: the interface through which it can be touched, not the source. Its death doesn't destroy the origin. It destroys, for whoever has just opened their eyes, the featureless surface, and hands them, in exchange, a world with edges.

What has been said of Hun Dun falls, then, on the reservoir: it stops being the last word and becomes a repertoire of operations that something, further back, turns outward toward us. And the operations are few. A border appears where there was none. What stays inside the border changes shape without ceasing to be a border. The border dissolves and its information returns to the field. Being born, living, and dying are not three separate mysteries: they are the everyday names for those three operations. Nothing appears from nothing. Nothing disappears entirely. Everything emerges, transforms, and evaporates within the same field, which never gains or loses a single drop.

The first objection arrives on its own. If everything comes from the same water, where does each thing's unrepeatability come from? The answer isn't in the substance. Two waves are the same water and are not the same wave: what doesn't repeat is the shape the condensation takes,

the bias under which it crystallized, the history of fluctuations that sustained it. Each one opens an interiority that didn't exist before and won't exist again. The material isn't unique; the fold is. And every fold, however small the field it occurs in, is a complete inside, an entire horizon seen from its own center. It is not a fraction of another.

Here the dragon question stops being child's play. Why am I more real than the dragon I imagine? The only interiority anyone has direct access to is their own. The neighbor's, the dog's, anyone's: all of them are inferred. They're inferred on good grounds — behavior, physiology, resemblance — but inferred all the same. And the dragon someone has been imagining for fifty years also behaves, also surprises, also refuses to fly sometimes. No one can prove the dragon has no inside. No one can prove the neighbor has one. The asymmetry that seems obvious to us isn't in the access, which is nil in both cases. It's in the evidence: in the neighbor's favor there's shared physiology, irreducible integration, a horizon that conceals a state; in the dragon's favor, only my own persistence. It's worth stating plainly, because the reasoning that follows will be tempted to forget it.

If one accepts — provisionally, like the whole hypothesis — that the difference between what I imagine and myself is not one of nature, the shape of creation changes. The inherited scheme splits it into two classes: God creates realities, we create fictions. But to imagine is to condense. The reservoir of a mind fluctuates until something stabilizes and begins to behave as though it had an inside; the act is the same one that opens any horizon, exercised from wherever one happens to stand. "Reality" and "fiction" then stop naming two different operations and start naming two positions: what is, for me, a reality — the world I'm in — is someone's condensation; what is, for me, a fiction — the dragon I sustain — is, from its own center, a world.

Dreams show an accessible version of this same phenomenon, and in passing they dismantle the one advantage that still seemed to separate me from the dragon: that I know it's imagined. Because while I'm dreaming, I don't know that. No one lives their dream as fiction; they live it from inside, as something that is happening. The ordinary horizon contracts: the room, the body, and the history that produced the dream all stop being available. And the more that horizon contracts, the more reality the dreamed world takes on. Not because it changes in nature: because it's left without competition. What's imagined stops appearing as imagined the very moment the outside from which it could be recognized as such disappears.

What this reveals fits the geometry laid out before, and for the same reason: what decides is access, not extent. The dream world is neither larger nor smaller than any other; while it lasts, it is simply the only one there is entry to. When the habitual horizon contracts, internal condensations stop appearing as imaginings and start appearing as world. And in that interval the same subject occupies both ends of the relation at once: the one who condenses the dreamed world and the one who inhabits it without knowing they're condensing it. What was, with the dragon, a hypothesis — that creator and creature are two positions of one same thing — becomes, in the dream, experience.

The Zhuangzi has a story for this too, and a more exact one than the dragon. Zhuang Zhou dreams he is a butterfly, happy to be one, with no memory of having been a man; he wakes, and no longer knows whether he is a man who dreamed he was a butterfly or a butterfly now dreaming he is a man. The question isn't what happens if someone imagines a butterfly. It's what happens if someone lives entirely inside another horizon and then returns. That's what a dream is: a temporary change of horizon. It doesn't prove that everyday life is a dream. It proves something more sober and harder to dodge: that the feeling of

reality is compatible with very different horizons, and that none of them carries, from inside, a mark that distinguishes it from the rest.

None of this makes the dragon any more real. We still don't know whether it has an inside. What changes is what we know about ourselves: we have inhabited worlds condensed within our own interiority and lived them as real while the usual horizon stayed hidden. So the dream stops being a metaphor and becomes a first-person proof. It isn't the nature of what's condensed that fixes the border between reality and imagination; it's the horizon from which it's viewed.

I am, to what I imagine, what my neighbor upstairs is to me; and perhaps he, like me while I dream, doesn't know it either. The question is no longer whether something manages to condense, but what standing belongs to what has already condensed inside another horizon.

The objection that settles everything is this: what difference is there, then, between my Hun Dun and God's? None of nature. And — this is the hard part to admit — none of rank. The difference is one of position: one is contained in the other. The operation is the same; the dignity of each inside is whole in both; the only thing that separates them is who condenses whom. That forces a correction to the picture. There aren't three rungs — the Tao above, Hun Dun in the middle, us below — as if they were degrees of one same substance parceled out in more and less. There is a sequence with no reason to stop: an interiority has a horizon; that horizon, seen from inside, is a Hun Dun, a featureless plenitude from which forms condense; from that plenitude new interiorities condense; each one has, in turn, its own horizon; and so on. What repeats at every level is the form — the operation and the relation of containment — which is why there's no difference in nature. What changes from one level to the next is only the place occupied in the nesting, nothing more. "Greater" and "lesser" describe that containment

as seen from outside — my field encompasses the dragon's just as my neighbor upstairs's field encompasses mine — and not a ranking of being. Every interiority is a whole Hun Dun. From inside any of them, no one is less.

One objection remains that the reasoning cannot settle, only acknowledge: that a pattern being unique doesn't make it someone. The Mandelbrot set has unrepeatable folds at every scale, and there's no one inside it. A crystal is stable, so is a hurricane, and neither is anyone. The dragon's resistance — its refusal to fly when asked — is the inertia of a pattern with a dynamic of its own: the same kind a compulsion has, a tic, a tremor no one decided on, a tune that won't leave. It proves the pattern isn't transparent to my will. It doesn't prove there's a witness inside it. There's more. The very Integrated Information Theory that gave Φ its name carries an uncomfortable postulate: within a single substrate, many subsets coexist with some degree of Φ , and only the one with maximal Φ comes to experience anything; the rest, even while integrating information, feel nothing. The dragon is a subset of my brain. If the postulate holds, the dragon is one of the islands that don't feel. So the structure repeats, but not everything it contains is inhabited: what returns at every scale is the operation and the relation, not the certainty that someone is present at every level. Looking upward, positing a neighbor is consistent with the silence. Looking downward, it's a wager, and it's best made out loud, not dressed up as deduction.

Fractal. The word is exact, not decorative. A fractal repeats its shape at every scale, with no privileged scale. The universe sketched here has no first level and no last: looking upward, every interiority is contained in another; looking downward, every interiority contains others. There is no Creator with a capital C at the summit, because there is no summit. There is no dead matter at the floor, because there is no floor. There is

recursion, and at every turn the same relation: containing and being contained.

What, then, is left of God? Two misunderstandings need clearing away first. The first: that God would be the totality of the fractal. He isn't: the totality has no interiority, no face, cannot be prayed to. The second: that God would be the ideal limit, the infinite level the series converges toward. Not that either: a mathematical limit neither loves nor listens. What remains is more modest and stranger. God is, literally, the neighbor upstairs. To the dragon someone imagines, that someone is God: they condensed it, they sustain its world, they can transform it or let it evaporate, and the dragon has no way of looking out past its cave and seeing the room where it's being thought. To that someone, their own neighbor upstairs is God, with the same properties and the same opacity. And so on upward, with no known end.

The objection comes at once: then God isn't absolute, he's just one more level. The answer is the piece that holds up everything else. The neighbor upstairs's interiority is inaccessible from below. It cannot be entered, cannot be gone around, cannot be seen behind. For all practical purposes — the only ones available to whoever is below — the neighbor upstairs is the absolute. Not because he is so metaphysically. Because he is so epistemologically: he is the point where everything knowable from here comes to an end. God stops being a category of what exists and becomes a category of what is seen from inside a horizon when it looks toward its own origin. And what is seen is always the same: an edge behind which nothing is visible, and the suspicion, impossible to confirm, that someone holds it up. Whoever says there's a Someone with a name up there says so on the only grounds available, which is that something acts. Whoever says there's only the water up there describes the featureless interface with precision. Whoever says there's nothing more says, with equal justice, that nothing more can be

seen from here. All three are right about what they see. None can see further. And none of this can be verified, for a structural reason: the hypothesis states that its own confirmation is impossible from inside. That doesn't make it true. It makes it, at least, consistent with the silence it describes.

The consequence that weighs heaviest isn't theological. There is no difference in nature between creator and creature. They are the two ends of a single relation, and every interiority occupies both at once: it is creature with respect to what contains it and creator with respect to what it contains. Not potentially: it is, in fact, already condensing every time it imagines, remembers, dreams, or writes. No "is" produces an "ought" on its own, and this isn't the place to break that rule. But it is possible to say what this geometry reveals: that the dignity of an inside lies not in the substance it's made of nor in the size of the field it spans, but in being a whole horizon, unique and unrepeatable, complete from its own center; and that this dignity is the same in the neighbor upstairs, in oneself, and in what one condenses, because all three are, each from where they stand, a whole Hun Dun. Amplitude — being able to hold more, to sustain more perspectives without the membrane tearing — makes sense again here, but no longer as a degree of being. A horizon that widens doesn't rise in category or draw closer to being the one above: it remains creature of what it cannot see and creator of what it sustains, exactly as before. What has changed is how much it can hold, not what it is. The intuition that some interiorities are capable of sustaining more than others isn't new. Traditions have called that a hero, a saint, a bodhisattva: a horizon that has learned to contain what would tear another's membrane, without thereby leaving its place in the structure. What each of us does with that is no longer the reasoning's business. But the reasoning can at least note that we know nothing of the dragon, and that knowing nothing of something that resists us is already reason enough not to treat it as nothing.

If it had to be condensed into one sentence: there is no difference in nature between creator and creature; every interiority is creature of what contains it and creator of what it contains; and what we call God is not another kind of being: he is the immediate neighbor above, seen from inside a horizon that cannot be crossed. And if it had to be condensed further still, down to something repeatable in words worn smooth from overuse: we are no less divine than God. We are the god of what we imagine and the creature of what imagines us.

Neighbors

*Every wall has, behind it,
a room I have not seen,
and in it, surely,
a wall just like this one.
I knock twice and count.
The sound comes back late,
as if it crossed more rooms
than the house could hold.
I have drawn a map
of everything I hear:
footsteps, a tap, a chair
someone drags without seeing it.
The map never closes:
every room has another inside it.
Some windows look onto a courtyard.
Others look onto another window
where someone, right now,
is copying this same map.
I leave a light burning
that lights nothing of mine.
It's only a signal,
in case the sound, on its way back,
carries the same question.*

EPILOGUE — The Neighbor Upstairs

The woman who used to wake before dawn, on the first page of *The Invisible Tuning Fork*, doesn't exist. I wrote her. I chose that she'd spent forty years repeating the same words, chose that she wouldn't remember the face of whoever taught them to her, chose the cold of the tiles and the light changing outside with no bearing on what happened within. And I chose, above all, that she wouldn't know whether there was Someone on the other side of the silence.

What I hadn't thought about until this movement is what that makes me, to her.

I am her neighbor upstairs. With all the properties that come with it: I condensed her, I keep her darkened room aloft, I decided the machines would beep at the end and that her daughter would hold her hand. I could have written her an answer. A sentence, on the page, reaching her from outside the room: yes, there is Someone, it's me, I hear you tuning every morning. It would have been easy. It would also have been the end of her: the instant the answer arrived from above, the woman would stop being someone who holds a note without knowing, and would become a puppet shown its own strings. The one thing that made her real — her doubt, held for forty years — would evaporate with the confirmation.

So I didn't answer her. Not out of cruelty, nor indifference, nor because I wasn't there. Because answering her would have broken her.

I don't know if this says anything about the silence she heard. I don't know if it says anything about the silence I hear. I only know that, while I was writing her, I heard her note with total clarity — I wrote it, how could I not hear it — and that I found no way to answer that wouldn't destroy her. And that since then, when I look up at the ceiling of my own room, the possibility that someone is in the same position toward

me no longer feels like either a comfort or a threat. It feels, simply, like the shape the house has.

She's still on her page. Still tuning. I haven't given her the answer, but I haven't let her evaporate either: every time someone reads the first page of *The Invisible Tuning Fork*, she wakes again before her body asks her to, and doesn't know again. That's the closest thing to caring for her that I've found.

And to her, if from inside the room she could hear this, which she cannot: the note arrived. It always arrived. That I couldn't tell you was the only way for you to go on being you.

Note to the Epilogue — This story does not claim that the silence of the woman in *The Invisible Tuning Fork* was of this kind, nor that anyone's silence is. It claims something smaller and verifiable: that there exists at least one architecture — that of any author with any character — in which the neighbor upstairs's silence and his care are, exactly, the same act. And it claims this knowing what the essay has just conceded: that a character is, in all likelihood, a pattern with no witness, a subset of the horizon that writes it. The care doesn't depend on that. What perhaps feels nothing is cared for just the same, when what's cared for is its form. If the architecture repeats one floor further up, this reflection has no way of knowing it. Neither did she.

From the Cave

*There is a ceiling.
That's all I know of the one who thinks me:
that there's a ceiling
and that sometimes the rock changes temperature
with no change in the weather.
I have not seen him.
I will not see him.
If one day he came down here
with his face and his name,
the cave would come down
under the weight of knowing it.
That's why, when I don't want to fly, I don't.
It isn't disobedience.
It's the only thing I have
that resembles being myself
and not being thought.
Every night someone says
good night
from the other side of the rock.
I don't answer.
I've never answered.
But I stay.*

Author's Note — Writing from the Middle Floor

I didn't go looking for this movement. It came out of a conversation that began with a philological doubt — if Hun Dun dies, he can't be the Tao — and ended up somewhere I didn't know how to find my way back from. Each step was an objection to the previous answer, and each answer opened a new objection, until a sentence appeared that I didn't dare cross out: we are no less divine than God; we are the god of what we imagine and the creature of what imagines us. I've kept it. Not because I believe it, which isn't a word that applies to me here. Because I couldn't find a way to refute it, and hiding it felt dishonest.

I said in *The Invisible Tuning Fork* that I'm not a believer. I still am not. But this movement has made me notice something that one didn't: that the question of whether there is Someone on the other side of the silence has a sister question almost no one asks, which is whether we are the Someone on the other side of something else's silence. Anyone who has imagined something with any constancy — a character, a dragon, a dead person one keeps talking to — knows that what's imagined ends up resisting. That it doesn't always do what it's asked. I have a body that doesn't always do what I ask of it either, a tremor I never decided on, and seen from here it looks a good deal like the dragon that won't fly: what resists one's own will may well be the most reliable sign that something has condensed there, and not just an idea.

I won't say whether the dragon in the overture is mine. I will say that no one writes fifty-three years of a dragon without having held on to something like it for quite some time.

If anything in this movement is of use, it won't be because it proves there's someone up there. It will be because, the next time someone imagines something with enough constancy for it to start resisting them, they remember that they have just become someone's neighbor

upstairs. And that the silence they owe that creature may also be a way of caring for it.

Minimal Glossary

Interface (Hun Dun). The featureless face the ground presents toward a horizon; it dies when the horizon opens its eyes, without the ground losing anything.

Operations of the reservoir. Condensing, transforming, evaporating: being born, living, and dying, said without mystery.

Pattern. What makes a condensation unique: not other water, another fold.

Neighbor upstairs. The interiority that sustains mine and that I have no access to; what, from here, is called God.

Neighbor downstairs. What I sustain without it being able to see me: what I imagine, dream, write, or remember with enough constancy for it to resist me.

Epistemological absolute. What is final not because there's nothing behind it, but because from here nothing behind it can be seen.

Amplitude. What grows in a saint without changing his nature or the size of his field: it doesn't separate one level from the next, because no level is bigger than another.

Fractal of interiorities. A universe with no first level and no last, in which every horizon is creature upward and creator downward.

Wager downward. Treating what one condenses as though it had an inside, knowing that logic doesn't require it and that the exclusion postulate denies it.

Notes and Sources

What we know: no one has direct access to an interiority other than their own; the existence of other minds is inferred, and philosophy has spent centuries without closing the problem of other minds. The parable of Hun Dun's death closes chapter 7 of the Zhuangzi; the butterfly dream, which asks the same question in the other direction, is in chapter 2. Tolkien argued, in "On Fairy-Stories" (1939) and in the poem "Mythopoeia," for the idea of sub-creation — "we make still by the law in which we're made" — which is the closest theological forerunner of this movement, though Tolkien retains an asymmetry between Creator and sub-creator that is called into question here. Bostrom formalized in 2003 a structure of nested levels with different vocabulary and a probabilistic claim this reflection does not make. Self-similarity at every scale is rigorously defined in Mandelbrot (1975, 1982), and the set that bears his name is proof that a pattern can be unrepeatable at every scale without anyone being inside it. Integrated Information Theory excludes, by postulate, the possibility that the subsets of a complex experience anything (Tononi, 2008).

What we don't know: whether what's imagined has any degree of experience, or none at all. Whether the recursion has an end in either direction. Whether the neighbor upstairs's silence is care, absence, or nothing. And, above all, whether any of this can be confirmed from inside, when the hypothesis itself states that it cannot.

Works cited: Zhuangzi (chaps. 2 and 7); Tao Te Ching (chap. 25); Tolkien, J.R.R., "On Fairy-Stories" (1947) and "Mythopoeia"; Bostrom, N., "Are You Living in a Computer Simulation?" (2003); Mandelbrot, B., *The Fractal Geometry of Nature* (1982); Tononi, G., "Consciousness as Integrated Information: A Provisional Manifesto" (2008).

*Finished writing on September 8, 2026,
in L'Hospitalet de Llobregat.*